X68000 Service Manual (CZ-634C / CZ-644C)

English translation from the Japanese original

Contents

SHARP Service Manual No. CZ-134

X68000 Personal Computer CZ-634C-TN CZ-644C-TN

Distribution targets: Sharp Engineering (Co.) Ltd. SS, SB.

Issued: April 1991

Contents

(Note at the bottom) ● Prompt and reliable service leads to repeat business. ● Proper maintenance connects you with customers. Ensure trust… Issue without fail.

Sharp Corporation Electronics Systems Division, Product Production Center

1. Hardware Configuration

1-1. Features

1) CPU Peripheral

- Uses 68000 (16.67 MHz), a 16-bit MPU.
- Capable of directly addressing up to 16MB of address space.
- Memory-mapped I/O (standard 2MB main memory equipped).
- Uses DMAC for 63450 and MFP for 68901.
- Multiple catalog ICs are supported.

2) Text VRAM and Graphics VRAM use bit-mapping methods.

- Screen resolution of 1024x1024 dots (supports display screens of 512x512 dots for graphics)
- Display resolutions can be selected from 768x512, 512x512, and 256x256.
- Supports both high resolution (31.5 kHz) and low resolution (15.98 kHz).
- In graphics mode, supports a color palette of 65536 colors, with 65536 selectable colors from the color palette (512x512 dots at selection time).
- In graphics mode at 768x512 dots, up to 16 colors from the 65536 colors can be arbitrarily selected per dot.

4) Smooth scrolling is possible in single dot units.

5) Sprite Features

- Can generate up to 128 sprites of a 16x16 dot pattern (maximum of 256 images).
- Up to 32 sprites can be displayed simultaneously in one line.
- Can display up to 128 sprites simultaneously on one screen.

6) Background switching capability.

7) Equipped with a function to prioritize sprites over the background in graphics mode.

8) Transparency specification, equipped with a semi-transparent sprite function.
9) Equipped with a superimpose function that uses high-resolution overlaid images (supports interlace method for reduced flicker).
10) CGROM includes ANK characters; JIS Level 1 and Level 2 Kanji are equipped as standard.
11) Equipped with FM sound source and voice-synthesis functions.
12) Equipped with a SCSI interface to support optical disk drives, such as CDROM as the next media, as well as analog RGB/I/F, RS-232C I/F, printer I/F, joystick I/F, mouse I/F, and other types of I/F.
13) Uses a cylindrical stepper motor type keyboard.
14) Equipped with 2 units of 5-inch floppy disk drives (2HD). Equipped with a trackball mouse.
15) Equipped with a 3.5-inch 80MB hard disk (CZ-634C-TN model comes with an internal optional hard disk).
16) SRAM Initialization Method

A simple function to initialize SRAM has been added to this machine. Without booting the OS, pressing the `CLR` key leads to a quick reset, and a message appears on the screen prompting for initialization. Press the `Y` key to initialize, or `N` key to cancel. This causes the SRAM to start in initialized state.

Note: Specifications and external appearance are subject to change without notice.

CZ-623C to major changes

■ Gate Array iX1197CE (OHM-2) ...... changed to iX1748CE (ASA) iX1099CE(MESSIAH) ...... changed to iX1749CE (DOSA) ■ Inhibit indicator connector addition ■ MPU HD68HC000PS10 ...... changed to MC68HC000B16 ■ FPU IC socket added ■ MPU clock can be set to two modes: 16.67MHz/10MHz. ■ 4M x 2 ROM iX1614CE (EVEN) ...... changed to iX1775CE iX1615CE (ODD) ...... changed to iX1776CE ■ BIOS ROM replacement IC socket can be fitted on the 2MB expansion RAM module (CZ-6B2EA). ■ The SCSI connector can only be connected to SCSI-specification devices. (Non-SCSI devices such as the CZ620H cannot be connected.) ■ SCSI connector power terminal has a fuse to prevent overcurrent, with a 1A fuse installed. Do not use any fuses other than the designated ones.

About service correspondence for print substrate modules For electronic controlled substrates, print substrate modules are subdivided as shown below. Please handle repairs by the respective method.

Part name	Distribution code	Service response method
Main substrate unit		Response by unit parts exchange in the substrate
FD controller substrate unit		
Control substrate unit		
I/O substrate unit		
Power LED substrate unit		
FD-LED substrate unit		
Video substrate unit		
Analog substrate unit		
Keyboard unit		
SCSI controller substrate unit	(CZ-644C)	

Part name	Distribution code	Service response method
SCSI controller substrate unit	(CZ-634C)	

1-2. Specifications

Item	Category	Name & Type	Conten
CPU	MPU	MC68HC000	16-bit M
	Sub CPU	MSM80C51	Keyboa
	DMAC	HD63450	4-chann
	FPU	MC68881	Floatin
	MFP	MC68901	Multifu
	CRTC	IX1093CEZZ (VICON)	Text/gra
Peripheral	LSI		
Sprite Controller	iX0906CEZZ (CYNTHIA)	Sprite function	
FDC	µPD72065	Controls built-in 5-inch 2HD FDD	
Video Controller	iX1095CEZZ (VIPS)	Display priority function. Special effects function	
SCSI Controller	MB89352	SCSI control	
SCC	Z8530	Serial communication controller. Dual-channel (RS-232C, mouse)	
RTC	RP5C15	Real-time clock	
FM Sound Source	YM2151	8-channel FM sound source generation	
PCM	MSM6258	Adaptive Differential PCM	
Synthesis PPI	µPD8255	Joystick support. Sound source switching control	
I/O	iX1604CEZZ	Floppy disk, peripheral IC decoder	
	iX1748CEZZ	Memory controller (ASA)	
	iX1749CEZZ	System controller (DOSA)	
	iX1094CEZZ	Video data selector	
	iX1096CEZZ	Video clock controller	

Note: CEZZ appears to be suffix for LSI chip codes, specifics can be machine dependent.

Item | Category | Name/Type | Content | Remarks

Memory

ROM | CG ROM (combined IPL ROM) | 1M byte (JIS first and second standards) | - |

RAM | Main Memory | 2M bytes (Standard) | Can be expanded to 12M bytes (Selectable up to 6M bytes with inbuilt connector, in 2M byte units.) |

Text VRAM | Bit Map Method | 1024 x 1024 Dot | 4 Planes |

| | 512K x 4 Bit | - | Dual Port Type, for DRAM |

Graphics V-RAM | Bit Map Method | 512K bytes x 1024 x 1024 Dot | 4 Planes | Dual Port Type, for DRAM |

| | (512 x 512 Dot | 16 Planes) | - |

Sprite V-RAM | 32K byte | - | - |

S-RAM | 16K byte | - | - |

Media Storage

- Inbuilt 3.5-inch hard disk 80M byte (Optional for CZ-634C)
- Expansion floppy disk drive for disk expansion

Interface

- Floppy Disk Interface: -
- SCSI Interface: -

- Keyboard Connector: For dedicated keyboards.
- CRT Interface: For analog RGB output.
- TV Control Connector: For dedicated display TV control.
- RS-232C Interface: 1 channel RS-232C.
- Mouse Interface: For attached trackball mouse.
- Printer Interface: Centronics standard format.
- Joystick Interface: Atari standard format (2 connectors).
- Audio Input/Output Connector: Line in/output, headphone output.
- Video Input Connector: For optional color image unit.

Additional Connections

- Other: EXPWON, VHT
- Expansion Slots: 2 Slots
- Power Supply: AC 100V
- Frequency: 50/60Hz
- Power Consumption: CZ-644C: 46W, CZ-634C: 41W

Page 5

Item Classification Name-Category Content Remarks Display Surface Text Screen 1024 x 1024 dots 4-plane Bitmap Method Size Graphics Screen 1024 x 1024 dots 4-plane Bitmap Method

```
                                                               (512 x 512 dots  16-plane)

           Text Surface       High-Resolution Mode          768 x 512 dots
                                                               512 x 512

                                                               256 x 256 (2-plane reading)

                             Low-Resolution Mode          256 x 256
                             (Overscan)                   512 x 512
                             (Interlaced)

           Display Surface    High-Resolution Mode          768 x 512 dots
            ↪  Per dot 65536 colors
```

1024 x 1024 of which any 16 colors can be specified

```
                                                               512 x 512

                                                               512 x 256 (2-plane reading)
                                                               256 x 256 (2-plane reading)

                             Low-Resolution Mode          512 x 256
                             (Overscan)                   256 x 256

                             (Interlaced)

                                                               512 x 512

           Graphics Screen    High-Resolution Mode
            ↪              Per dot, 65536 colors out of which any
```

16 colors can be s

Per dot, 65536 col

256 colors can be ⌋

Low-Resolution Mode

```
512 x 256 (Overscan) 256 x 256
      ↪
      ↪                        Per dot,
      ↪  65536 colors out of which any
  512 x 512 (Interlaced)
      ↪
      ↪                          16
      ↪  colors can be specified
```

512 x 512

256 x 256 (2-plane reading)

CZ-634C-TN CZ-644C-TN

Item Content

Smooth Scroll Function Text screens can be scrolled circularly in dot units. Graphics screens can be scrolled spherically in dot units.

Special Screen Control Function Graphics VRAM image input function, text raster copy function, graphics high-speed clear, text bit mask function.

Priority Function

- Priority order can be specified among text, graphics, and sprites.
- Priority order can be specified for each graphics screen when using 2, or 4, graphics screens in the 512 x 512 dot mode of the graphics actual screen.

Palette Function Any color can be switched instantaneously.

Half-Transparency Function Half-transparent display is supported.

Special Priority Function A function that can give the highest priority to any area of the graphics screen within the displayed screen.

Superimpose Function

- Low resolution overscan superimpose is supported. (Pseudo high resolution by interlace method is also supported)

Item Class Name/Type Content Sprite Sprite Pattern Definition Size 16 x 16 dot pattern

```
        Number of Definitions   128 patterns (BG0,1 unused)
            (up to 256 patterns)
        Colors  16 colors / 65536 colors per pattern
            (dot units)
            256 colors / 65536 colors for the whole screen
    Display Coordinate System   1024 x 1024 dots
        Display Screen  Horizontal 512 dots or 256 dots
            Vertical 512 lines or 256 lines
        Display Limit   128 sprites / screen
            32 sprites / line
```

1-3. Block Diagram

[Textual summary of the fold-out block diagram. Component names and clock values were read directly from the source page); the original is a large interconnected diagram that cannot be linearised exactly. The original text layer of this page was unrecoverable noise and has been replaced.]

Processor / bus

- MPU: MC68HC000 (16.67 MHz / 10 MHz, 2-mode)
- FPU: MC68881 (IC socket)
- DMAC: HD63450
- CG / BIOS ROM: 1 Mbyte
- SRAM: 64 Kbyte

Clocks

- Main oscillator: 40 MHz / 33.33 MHz; also 16 MHz and 4 MHz
- Video / CRTC dot clocks: 38.863632 MHz, 69.5519 MHz, 50.350 MHz
- Sprite clock: 32.209 MHz

Video

- CRTC
- TEXT VRAM 512 Kbyte ×2
- GRAPHIC VRAM 512 Kbyte ×2
- Sprite controller + Sprite VRAM 32 Kbyte
- Video controller (GVA), Palette, D/A → CRT

Sound

- OPM (YM2151)
- ADPCM (MSM6258)
- PCM-PAN mixer → Amplifier → headphone/speaker, Line out, MIC in

Peripherals

- RTC (RP5C15, 32.768 kHz crystal)
- PPI (8255): ports A / B / C; Joystick 1, 2
- SCSI (MB89352) → SCSI HD
- FDC
- SCC → RS-232C
- MFP (MC68901)
- System controller, I/O control bus

1-4. System Configuration

This diagram shows the system expansion centered around the main unit.

Main Unit

- X68000

```
CZ-63AC-TN
CZ-64AC-TN
```

Exclusive Display TV

Expansion Board

- Expansion RAM Board (2M)

```
CZ-6BE2 Expansion RAM (2M)
CZ-6BE2B
```

- Expansion RAM Board 2 (4M) CZ-6BE4

Universal I/O Card

- CZ-6BU1

GP-IB Board

- CZ-6BG1

Joystick Card

- CZ-8NJ1

Device RS-232C Card

- CZ-6BF1

Device Processor Board

- CZ-6BP1

Device Control Board

- CZ-6BP2

FAX Board

- CZ-6BC1

MIDI Board

- CZ-6BM1

Sound Amplifier Speaker

- AN-S100

Peripherals

- Intelligent Joy Unit Controller CZ-8NU2
- Mouse CZ-8NM2A
- Mouse Trackball CZ-8NM3

Network

- Module Unit Controller CZ-8TM2
- RS232C Cable

```
CZ-8LM1 (Programming)
CZ-8LM2 (Library Link)
```

- LAN Port

```
CZ-6BL1 (For Local Area Network)
LAN Kit
CZ-6BL2 (Including Adapter and TAP)
```

Imaging and Image Input/Output Switch

- Color Image Copier CZ-6VT1
- Color Image Scanner CZ-8NS1 GX-220X
- Video Capture Card CZ-6BV1

Printer

- 24-Dot Matrix Color/Monochrome Printer CZ-8PC3
- 48-Dot Matrix Color Printer CZ-8PC4

Display Section

- RGB System Selector CZ-6TU

File Section

- Removable Mass Storage Disc System (594MB) CZ-6MO1

Print Buffer

- Constant Print Buffer CZ-6EB1

System Rack

- System Rack CZ-6SD1

24-Pin Dot Impact (13 Color) CZ-8PG1

- 24-Pin Dot Impact (13 Color)

```
Portable Printer
CZ-8PG2
```

- Color Video Printer CZ-6PV1
- Color Image Printer I/O-735X I/O-735X-B

24-Pin Dot (13 Color) with Scanner CZ-8PK10

CRT Filter BF-6BP10 (14/15 inches)

2. Names of Each Part

2-1. Front of the Computer Body

1. Floppy disk drive access display lamp
2. 5-inch floppy disk drive
3. Eject button
4. Headphone jack (PHONES)
5. Joystick connector (JOY STICK 1)
6. Power switch (POWER)
7. Volume control knob (VOLUME)
8. Keyboard connector (KEYBOARD)
9. Mouse connector (MOUSE)
10. Indicator panel
11. Clock frequency switching switch (CLOCK SPEED)
12. Reset switch (RESET)
13. Interrupt switch (INTERRUPT)
14. Carrying handle

Page 10

2-2. Rear View of Computer Unit

1. Slot Cover (I/O SLOT)
2. Remote Terminal (REMOTE)
3. Printer Connector (PRINTER)
4. Frame Arm (FG)
5. Stereoscopic
6. Dedicated Stereo Display TV Control Terminal (TV CONTROL)
7. See-Through Color Terminal (SEE THROUGH COLOR)
8. Image Input Connector (IMAGE IN)
9. Analog RGB Signal Output Connector (ANALOG RGB OUT)
10. RS-232C Connector (RS-232C)
11. Audio Input Terminal (AUDIO IN)
12. Video Output Terminal (AUDIO OUT)
13. Joystick Terminal (JOY STICK 2)
14. Grounding Cord
15. Main Power Switch (POWER)
16. Service Connector (AC 100V OUT)
17. External Floppy Disk Drive Connector (EXPANSION FDD)
18. SCSI Connector

**CZ-634C-TN CZ-644C-TN

3. Hardware 3-1. Memory Map**

Key:

- CRTC: Cathode Ray Tube Controller
- VIDEO: Video
- DMA/C: Direct Memory Access/Controller
- NMI SET: Non-Maskable Interrupt Set
- INT: Interrupt
- R/W CTRL: Read/Write Control
- PRT IC: Printer Integrated Circuit
- FDD: Floppy Disk Drive
- HDC: Hard Disk Controller
- SCC: Serial Communication Controller (Z8530)
- 8255: Programmable Peripheral Interface (8255)
- INT: Interrupt
- FPU: Floating Point Unit
- SPRITE REGISTER: Sprite Register
- SPRITE VRAM: Sprite Video RAM
- EEPROM: Electrically Erasable Programmable Read-Only Memory
- USER I/O AREA: User Input/Output Area
- SUPERVISOR I/O AREA: Supervisor Input/Output Area
- MEMORY SW (RAM): Memory Switch (RAM)
- SYSTEM I/O: System Input/Output
- CG-ROM (128 KB): Character Generator Read-Only Memory
- IPL ROM: Initial Program Load Read-Only Memory
- MAIN MEMORY (USER AREA): Main Memory (User Area)
- GRAPHIC VRAM: Graphic Video RAM
- TEXT VRAM: Text Video RAM
- SUPERVISOR AREA: Supervisor Area (User Area)

Additional labels within blocks:

- 06B0000H: (Hexadecimal address), etc.
- 00FFFFFFH: (Hexadecimal address limit)

Note: The addresses provided (like 06B0000H, 00CFFFFF, etc.) appear to be hexadecimal memory addresses used in the context of the system's memory map.

3-2 I/O Port Address List

- Indicates invalid. W is WRITE only, R is READ only, and R/W is READ/WRITE.

Item	Port Address	Function	Remarks
C	E80000H	Horizontal Scroll	
R	E80002H	Horizontal Synchronization End Position	
T	E80004H	Horizontal Display Start Position	
C	E80006H	Horizontal Display End Position	
C	E80008H	Vertical Scroll (Y)	
R	E8000AH	Vertical Synchronization End Position	
W	E8000CH	Vertical Display Start Position	
E	E8000EH	Vertical Display End Position	
T	E80010H	External Synchronization Adjust	
H	E80012H	Raster Register Write Position	
W	E80014H	X Axis Scroll X	
E	E80016H	Y Axis Scroll Y	
W	E80018H	Screen 0 X	
A	E8001AH	Screen 0 Y	
H	E8001CH	Screen 1 X	
E	E8001EH	Screen 1 Y	
O	E80020H	Screen 2 X	
P	E80022H	Screen 2 Y	

Item	Port Address	Function	Remarks
L	E80024H	Screen 3 X	
E	E80026H	Screen 3 Y	
S	E80028H	Memory Mode; Display Mode Setting	
E	E8002AH	Text Access, High-Resolution Clear Plane	
R/W	E8002CH	Work Area Initialization Raster	
E	E8002EH	Bit Mask Register	
H	E80040H	CRT Control Register	
Gra	E82000H	16 Color Mode - 256 Color Mode	Graphics Palette
Pal	E82010H	or 65536 Color Mode	
let	E8210EH		
	E82020H		
	(B821FEH) R/W		
Tex&	E82200H	Text (Sprite Color Table 0) Common Palette	Text, Sprite Palette
SP	E8221EH		
Pal	E82210H		
let	(E82220H)	Sprite Color Table 1 Palette	Sprite Palette
	E8223EH		
	E82240H	Sprite Color Table 2 Palette	
	E8225EH		
	(E8230EH)	Sprite Color Table 15 Palette	
	E8232EH		
	E8233EH		
	E823F0EH)		
Video	E82400H	Memory Mode Setting	
Cont-	E82500H	Priority Setting	
roller	E82600H	Special Mode, Screen Display Control	Special Priority (Semi-Transparent)

CZ-634C-TN
CZ-644C-TN

No.	Port Address	Function	Remarks
D	E84000H	R/W	Channel Status Register
M	E84001H	R	Channel Error Register
A	E84004H	R/W	Device Control Register
C	E84005H	R/W	Operation Control Register
	E84006H	R/W	Sequence Control Register
	E84025H	R/W	Normal Interrupt Vector
	E84027H	R/W	Error Interrupt Vector
	E8402DH	R/W	Channel Priority Register
	E84029H	R/W	Memory Function Code
	E84031H	R/W	Device Function Code
	E84039H	R/W	Base Function Code
	E8400AH	R/W	Memory Transfer Count (Word)
	E8401AH	R/W	Base Transfer Count (Word)
	E8400CH	R/W	Memory Address Register (LongWord)
	E8401CH	R/W	Device Address Register (LongWord)
	E8401DH	R/W	Base Address Register (LongWord)
	E840FFH	R/W	General Control Register
Area Set	E86001H	W	Supervisor Domain Setting
M	E88001H	R	GRIP Data Register
F	E88003H	W	Active Edge Register
	E88005H	W	Data Direction Register A
	E88007H	R	Interrupt Enable Register A
	E88009H	R/W	Interrupt Enable Register B
	E8800FH	R/W	Interrupt Pending Clear A
	E8800DH	R/W	Interrupt Pending Clear B
	E8800EH	R/W	Interrupt Service Level A
	E8800FH	R/W	Interrupt Service Level B
	E88010H	R/W	Interrupt Mask Register A
	E88011H	R/W	Interrupt Mask Register B

No.	Port Address	Function	Remarks
P	E88015H	R/W	Vector Register
	E88091H	W	Type A Control Register
	E88093H	W	Type B Control Register
	E8808FH	W	Type C/D Control Register
	E88071H	R/W	Type A Data Register
	E88081H	R/W	Type B Data Register
	E880E1H	R/W	Type C Data Register

Item	Port Address	Function	Remarks
M	E88025H	R/W	Timer D-Data Register
F	E88027H	R/W	Sync Character Register
P	E88029H	W	USART Control Register
	E8802BH	R/W	Receive Status Register
	E8802DH	R/W	Receive Status Register
	E8802FH	R/W	USART Data Register

| R | E8A001H | R/W | 1 Second Counter / ICKLOUT Select | | | T | E8A003H | R/W | 10 Second Counter / Adjust | | | C | E8A005H | R/W | 1 Minute Counter / Alarm 1 Minute Register | | | | E8A007H | R/W | 10 Minute Counter / Alarm 10 Minute Register | | | | E8A009H | R/W | 1 Hour Counter / Alarm 1 Hour Register | | | | E8A00BH | R/W | 10 Hour Counter / Alarm 10 Hour Register | | | | E8A00DH | R/W | Day Counter / Alarm Day Register | | | | E8A00FH | R/W | 1 Day Counter / Alarm 1 Day Register | | | | E8A011H | R/W | 10 Day Counter / Alarm 10 Day Register | | | | E8A013H | R/W | 1 Month Counter | | | | E8A015H | R/W | 10 Month Counter / 12-24 Hour Select | | | | E8A017H | R/W | 1 Year Counter / 10 Year Counter | | | | E8A019H | R/W | 10 Year Counter | | | | E8A01BH | R/W | Mode Register | | | | E8A01DH | W | Test Register | | | | E8A01FH | W | Reset Controller | |

| Printer | E8C001H | W | Printer Data | | | | E8C003H | R | Printer Strobe | | | | E9C001H | R | Printer Busy | |

| System Port | E8E001H | R/W | Contrast Adjustment (D/A) | | | | E8E003H | R/W | TV Control | | | | E8E005H | W | Video Input Control | | | | E8E007H | R/W | H/L LED Lighting, NMI Reset, Key Control | | | | E8E00BH | R | MPU Operation Clock Frequency Determination Port | | | | E8E00DH | W | SRAM Write Enable Control | | | | E8E00FH | W | POWER OFF Control | |

| FM Sound Source | E90001H | W | FM Sound Register / Address Port | | | | E90003H | R/W | FM Sound Register / Data Port | |

| Sound Synthesis | E92001H | R/W | ADPCM Status (IN) / ADPCM Command (OUT) | | | | E92003H | R/W | ADPCM Data Register (IN/OUT) | | | | E9A005H | W | ADPCM Output, Sampling Frequency Switching Register | 8255 Port C |

Item	Port Address	Function	Remarks
Floppy Disk	E94**1H R	FDCA Status Register (IN)	
	E94**3H R/W	FDC Data Register (IN/OUT)	
	E94**5H R/W	Drive Status (IN/Drive Control OUT)	Optional Signal
	E94**7H W	Access Drive Select, 2HD/2DD Switching (OUT)	
SCSI	E96021H R/W	Bus Device ID	
	E96023H R/W	SCSI Control	
	E96025H R/W	Command	
	E96029H R/W	Interrupt	
	E9602BH R/W	R...Re-Examine Status W...SCSI Controller Diagnosis	
	E9602DH R	Status	
	E9602FH R	System Status	

Item	Port Address	Function	Remarks
	E96031H R/W	Phase Control	
	E96033H R	Data Transfer Counter	
	E96035H R/W	Data Address	
	E96037H R/W	Temporary Register	
	E96039H R/W	Transfer Byte Count High	
	E9603BH R/W	Transfer Byte Count Mid	
	E9603DH R/W	Transfer Byte Count Low	
SCC	E98001H R/W	SCC Command Port B	
	E98003H R/W	SCC Data Port B	
	E98005H R/W	SCC Command Port A	
	E98007H R/W	SCC Data Port A	
Joystick	E9A001H R	Joystick 0	8255 Port A
	E98003H R	Joystick 1	8255 Port B
8255	E9A007H W	8255 Control Word Register	
FD, PR	E9C**1H R/W	FDC, FDD, Printer Interrupt Status (IN)	
	E9C**3H R/W	FDC, FDD, Printer Interrupt Mask (OUT)	
	E9C**5H R/W	FDC, FDD, Printer Interrupt Vector	
FPU	E9E000H R	Response Register	Internal Option
	E9E002H W	Control Register	
	E9E004H R	Status Register	
	E9E006H R/W	Restore Register	
	E9E008H R/W	Operation Word Register	
	E9E00AH W	Command Register	
	E9E00CH -	(Reserved)	
	E9E00EH W	Condition Register	
	E9E010H R/W	Operand Register High Word	
	E9E012H R/W	Operand Register Lower Word	
	E9E014H R	Register Select	
	E9E016H -	(Reserved)	
	E9E018H W	Instruction Address Register High Word	
	E9E01AH W	Instruction Address Register Lower Word	
	E9E01CH R/W	Operand Address Register High Word	
	E9E01EH R/W	Operand Address Register Lower Word	

CZ-634C-TN CZ-644C-TN

Item Port Address Function Remarks Expansion memory board / super visor setting port EAFF81H W Setting port for memory area 200000H~3FFFFFH Valid only when an option board is installed.

```
EAFF83H W   Setting port for memory area 400000H~5FFFFFH
EAFF85H W   Setting port for memory area 600000H~7FFFFFH
```

Sprite EB0000H R/W Sprite X coordinate

```
EB0002H R/W Sprite Y coordinate Sprite 0
EB0004H R/W Sprite control
EB0006H R/W Sprite priority

EB03F8H R/W Sprite X coordinate
EB03FAH R/W Sprite Y coordinate Sprite 127
EB03FCH R/W Sprite control
EB03FEH R/W Sprite priority

EB0800H R/W Background 0 X coordinate
EB0802H R/W Background 0 Y coordinate
EB0804H R/W Background 1 X coordinate
EB0806H R/W Background 1 Y coordinate
EB0808H R/W Background control
EB080AH R/W Horizontal total
EB080CH R/W Horizontal display
```

```
    EB080EH R/W Vertical display
    EB0810H R/W Resolution
```

Sprite PCG area EB0800H R/W Sprite 0 PCG Area

```
    EB807EH R/W
    EBBF80H R/W Sprite 127 PCG Area
    EBBFFEH R/W
```

Sprite text area EBC000H R/W Text area 0

```
    EBDFFEH R/W
    EBE000H R/W Text area 1
    EBFFFEH R/W
```

3-3. Area Set

In this book, the data can be set in the register of E86001H in the range from the beginning of the memory to 2MB (000000H - 1FFFFFH), allowing any location to be set as a supervisor area in 8KB units from the beginning of the memory.

- Register Port Address: E86001H, Write only 8-bit Register

```
D7  D6  D5  D4  D3  D2  D1  D0
       (Lower 8 bits are valid)
```

- Operation: In the range specification of 2MB from the beginning of the main memory, it becomes possible to specify the supervisor area by data settings.

```
    However, the setting is 8KB units (dividing 2MB into 256 parts).
    Example: Memory map when setting data is 7FH
```

000000H Supervisor Area 000000H Port Setting Value Port Data Setting Value Supervisor Area FFFFFFFFH 100000H Supervisor & User Area 0 0 ~ 1FFFFH (8KB)

```
                                                  1                                0 ~ 3FFFH
                                                  2                                0 ~ 5FFFH
                                                  3                                0 ~ 7FFFH
                                                  4                                0 ~ 9FFFH
                                                  5                                0 ~ BFFFH
                                                                                       .
                                                                                       .
                                                                                       .
```

200000H Supervisor & User Area 127 0 ~ FFFFFFH (1MB)

```
                                                                                       .
                                                128                              0 ~ 10FFFFH
                                                                                       .
                                                                                       .
```

FFFFFFFH 254 0 ~ 1FDFFFH

```
                                          255                              0 ~ 1FFFFFH (2MB)

             Supervisor & User Area
```

FFFFFFFH

- Expansion Memory Board Supervisor Setting Port By writing data into the supervisor area setting port, it is possible to specify the supervisor area in 256KB units for the memory area on the extension board.

Memory Area Supervisor Area Setting Port 200000 ~ 3FFFFFH EAFF81H 400000 ~ 5FFFFFH EAFF83H 600000 ~ 7FFFFFH EAFF85H (Supervisor area mode write only port)

Writing Data

```
D7  D6  D5  D4  D3  D2  D1  D0
       (Lower 8 bits are valid)
```

Note: H means hexadecimal in the context of memory and register addresses.

Writing data of the 2-bit lower digit corresponds to each 256K byte memory region. Writing 0 addresses it to Supervisor-only region, and writing 1 addresses to Supervisor & User region. The initial state is set to Supervisor & User region.
(Example) When EAFB| | HC-07 bit is written to the supervisor mode,
200000 to 2BFFFFH: Supervisor Mode
2C0000 to 3FFFFFH: Supervisor & User Mode

Memory Region (HEX)	Corresponding Bit
Memory Region 0	000000~03FFFF
Memory Region 1	040000~07FFFF
Memory Region 2	080000~0BFFFF
Memory Region 3	0C0000~0FFFFF
Memory Region 4	100000~13FFFF
Memory Region 5	140000~17FFFF
Memory Region 6	180000~1BFFFF
Memory Region 7	1C0000~1FFFFF

The above is valid when the optional 2M byte unit memory board is installed via the internal port connector. For example, in the case of expanding by 2M bytes, respond to each memory region 400000~5FFFFH and 600000~7FFFFH. Note: EAFP83H and EAFP85H may become inactive. Note) Models CZ-6BE2, 4/2/4M byte memory have the same port and function as the built-in ports.

3-4. System Ports

No.	Register Address	D07	D06	D05	D04	D03	D02	D01	D00
1	E8E001H	*			*				
2	E8E003H	*			*			3DL	3DR
3	E8E005H	*		*					
4	E8E007H	*		*	*			HRL	
5	E8E00BH	*	*		*	*	*	*	*
6	E8E00DH			SRAM Write Enable Control					
7	E8E00FH	*		*	*	*			*

● System Port Register Address Map (* is invalid) ● System Port Register Details

Bit	WRITE	READ
D00	3DR	3DR
D01	3DL	3DL
D02		FIELD
D03	TV Remote Signal	TV ON/OFF Status

1. E8E001H[READ/WRITE]
   - 16-level contrast adjustment for the computer screen (value from 0H[dark]—FH[bright]).
2. E8E003H[READ/WRITE]
   - D03 WRITE MODE: By controlling “0” and “1”, it can be used as a TV remote signal. In READ MODE, you can know the TV ON/OFF status (0 for TV ON, 1 for

TV OFF). However, it cannot be used simultaneously with the TV remote from the keyboard.

3. E8E005H[WRITE]
   - A system port for computer control of the optional ”Digital Tape Recorder”.
4. E8E007H[READ/WRITE]

Bit	WRITE	READ
D01	HRL	HRL Status
D02	NMI Reset	
D03	Key Ready	Key Jack Status

Page -20-

.D01 is usually used for switching the dot clock, so usually set it to “0.” Turn the NMI switch ON, and the MPU processes the highest priority interrupt (NMI interrupt). This NMI interrupt processing can be processed until the next clock interruption only when a reset is issued. So, if “1” is written to D02 after the CPU processing in the NMI interrupt routine, there is no need to reset this NMI. Also, if “1” is not written to D02, the NMI interrupt processing will not return, and the NMI switch will not function.

.D03 is used for WRITE MODE. During this mode, data can be sent from the CPU80C51 on the board to the MFPR (terminal). The written data cannot be sent to the keyboard or the printer when “1” is written. In READ MODE, you can know whether the keyboard jack is connected to the keyboard by seeing if it is in a floating state (”1” when there is no connection and ”0” when there is a connection).

5. E8E00BH[READ]

This port is used to determine the current operation mode of the MPU by setting it to Read Only. Among the 8-bit data read to the port, only the value of bit 0 is valid (all other bits remain “1”). When bit 0 is “1,” the MPU is in 10MHz operation mode, and when bit 0 is “0,” the MPU is in 16MHz operation mode (MPU operates at a clock frequency of 16.67MHz).

Note 1: For the 10MHz operation mode of CZ-600C series only, if this port is accessed, all bits other than bit 0 will remain “1.”

6. E8E00DH[WRITE]

Usually used for SRAM. It is set to Read Only, protecting the SRAM contents from being destroyed due to program runaway. When “1” is written, SRAM Write Enable is set, and by writing any other value, it will be in Read Only mode.

7. E8E00FH[WRITE]

When entering 00H --> 0FH ---> 0FH in that sequence, the power (Vcc1 OFF) will be turned off. Any other code sequence will be invalid. By doing so, it ensures that the power is not easily turned off.

3-5 Interrupts

< MPU68000 Interrupts >

Level	Source	Description	Reason
High 7	NMI	Interrupt by external NMISW etc. (auto vector interrupt)	Auto vector inte
High 6	MFP	Various timers, KEY data reception, H-SYNC, V-DISP interrupts (vector interrupt)	Vector interrupt
5	SCC	RS-232C, mouse data reception interrupt (vector interrupt)	Vector interrupt
4	Expansion	Expansion I/O slot	-
3	DMAC	Interrupts by data transfer etc. (vector interrupt)	Vector interrupt
2	Expansion	Expansion I/O slot	-
Low 1	Floppy	FDC, FDD, hard disk, printer BUSY etc. interrupt	Printer (FDC > F

< MFP (Multi-Function Peripheral) Interrupts, read-out ports >

Priority	Channel	Description	General Name
High 15	1111	H-SYNC signal from CRTC	General Purpose Interrupt 7(7)
High 14	1110	IRQ signal from CRTC (any raster selectable)	General Purpose Interrupt 6(6)
13	1101	V-DISP signal from CRTC	Timer A
12	1100	KEY data reception completion	Receiver Buffer Full
11	1011	KEY data reception error	Receive Error
10	1010	KEY data transmission completion	Transmit Buffer Empty
9	1001	KEY data transmission error	Transmit Error
8	1000	USART (keyboard) serial clock	Timer B
7	0111	CLKOUT signal from RTC (1Hz)	General Purpose Interrupt 5(5)
6	0110	V-DISP signal status from CRTC	General Purpose Interrupt 4(4)
5	0101	8-bit general purpose timer (input clock 4MHz)	Timer C
4	0100	8-bit general purpose timer (input clock 4MHz)	Timer D
3	0011	Interrupt detection from FM sound source	General Purpose Interrupt 3(3)
2	0010	ON/OFF detection by POWER switch	General Purpose Interrupt 2(2)
1	0001	ON/OFF detection by EXPWON signal from expansion I/O slot	General Purpose Interrupt 1(1)
Low 0	0000	ON/OFF detection by ALARM signal from RTC	General Purpose Interrupt 0(0)

Breakdown Register Address D07 D06 D05 D04 D03 D02 D01 D00 MFP E88017H Set

```
0 - Automatic interrupt end mode
1 - Software interrupt end mode
```

SCC (Channels A and B shared) Set in Register 2 for write

DMAC E84025H Set (Channel 0) (Normal Interrupt Vector) (Internal 2HD) E84027H Set (Error Interrupt Vector)

DMAC E84065H Set (Channel 1) (Normal Interrupt Vector) (Hard Disk) E84067H Set (Error Interrupt Vector)

DMAC E840E5H Set (Channel 2) (Normal Interrupt Vector) (Memory) E840A7H Set (Memory) (Error Interrupt Vector)

DMAC E840A5H Set (Channel 3) (Normal Interrupt Vector) (Audio Synthesis) E840E7H Set (Error Interrupt Vector)

Floppy E9C003H Set ·Printer FDC Interrupt 0 0

```
FDD Interrupt      0 1
Hard Disk Interrupt  1 0
Printer Interrupt   1 1
```

(Note: ”MFP,” ”SCC,” ”DMAC,” ”FDC,” ”FDD,” and the like typically refer to specific hardware components or functionalities in computing, e.g., Multi-Function Peripheral, Serial Communication Controller, Direct Memory Access Controller, Floppy Disk Controller, Floppy Disk Drive, etc.)

3-6. IPL

- IPL ROM address: FC0000H - FFFFFFFH (256 KB)
- Access:

```
1. Supervisor program, data area
2. Read-only
```

Reset Sequence:

1. Upon reset, part of IPL ROM 1 (F******H) appears in the top 64 KB of the memory map (000000H-00FFFFH).
2. At the beginning of the IPL ROM 1, there are two long words that store the program counter and stack pointer values for the 68000 MPU reset exception process.
3. Once the reset signal is removed, the MPU fetches the long word data from the ROMs shown at the top of the memory map.
4. At this time, the MPU executes instructions based on the program counter. Meanwhile, the IPL ROM 1 image disappears, and the system goes into the state shown in figure 3-2.

   Note: The value of the program counter stored in the IPL ROM 1 must point within the address range of IPL ROM 1. This is because the hardware is designed so that accessing addresses (FF0000H-FFFFFFFH) makes the IPL ROM 1 image disappear from the top of the memory map. The ROM image appears in the memory map upon power-on reset or manual reset (Figure 3-1). (The ROM image does not appear for the 68000 MPU reset command execution.)

Diagram 3-1 Diagram 3-2 | FC0000H ---- | | IPL ROM1 ----------------| ------- IPL ROM1 | | 00000H | --| | 00FFFFEH | Main Memory

4. Screen Configuration and Control
4-1. Screen Configuration

This machine has three independent screens: text, graphics, and sprite. Control of the text screen and graphics screen is carried out by the CRTC, and control of the sprite screen is carried out by the sprite controller. Additionally, various functions such as text screen, graphics screen, sprite screen priorities, .transparency, special priority functions, window functions on each screen, and screen blanking functions are controlled by the video controller.
When accessing the screen, set the CRTC and sprite controller numbers regardless of whether they are used or not.

(Figure showing screen control block diagram)

1) Text Screen (Bitmap Method)

- Screen used for displaying ANK characters and kanji characters
- Similar to the graphics screen setting of the conventional X1 & X1turbo series, the display data direction is horizontal
- Scrolling of the text screen is cylindrical scrolling
- As a display mode, it supports both high resolution (horizontal scan frequency 31.5kHz) and low resolution (horizontal scan frequency 15.98kHz)

2) Graphics Screen

- A screen for performing graphics processing such as line drawing or painting.
- The display data settings are performed in the background.
- The scrolling of the graphics screen is a dual screen scroll.

- As a display mode, it supports high-resolution (horizontal sync frequency 31.5kHz) and low-resolution (horizontal sync frequency 15.98kHz).

[Illustration] Top Display Screen Left Right Bottom

3) Sprite Screen

- A screen for smoothly moving sprite patterns dot by dot, as used in NES or MSX.
- As a display mode, it supports high-resolution (horizontal sync frequency 31.5kHz) and low-resolution (horizontal sync frequency 15.98kHz).

4-2. Control of Text Screen and Graphics Screen (CRTC) This CRTC has the capabilities to support the dual port DRAM (MB81461) used in the text and graphics VRAM and has internal registers unique to the custom CRTC-23, providing the following screen control functions: (1) Generation of horizontal and vertical sync signals (2) Generation of display start and display timing signals (3) Scrolling of text and graphics screens (4) Automatic data transfer functions to graphics VRAM (5) Line copy and bit mask functions for text VRAM (6) Display single/double switching function, simultaneous access, and switching function (7) Graphics VRAM high-speed access function (8) External sync signal control function (at superimpose time)

Note that it can temporarily interrupt the CPU access to VRAM, depending on the need for V-DISP signal period (at MFPG0 GPIP4*=”0”). Moreover, concerning DRAM's SAM (Serial and Control Merge) refresh cycle, it is controlled by the CRTC during the horizontal blanking period.

Page 26

● CRTC Specifications and Internal Registers

Table 4-1 CRTC Specifications

Summary	High Resolution	Low Resolution
Display Mode	Non-Interlace	Interlace

	Horizontal (kHz)	Horizontal (usec)	Vertical (Hz)	Vertical (msec)
Synchronous Frequency	31.5		55.46	
	22.09	16.25	55.46	16.25
Data Display Period (1)	31.75	52.69	16.25	16.270
Synchronization Period (2)	31.75	3.45	18.03	0.191
Synchronization Retrace Width (3)	3.45	4.14	0.191	1.111
Back Porch (4)	4.14	4.94	1.111	0.876
Front Porch (5)	2.07	1.65	0.476	0.187

[Diagram]

(1) Video Signal (2) Synchronization Signal (3) Front Porch (4) Back Porch (5) Synchronization Retrace Width

- Video Signal: Analog 0.7Vp-p (75Ω termination) positive polarity
- Synchronization Signal: TTL level negative polarity

Note: The numbers `(1), (2), (3), (4), (5)` correspond to the parts of the diagram provided below the table in the original document.

Special CRT Controller Features (Text) The text screen has the following features: (1) Scroll Performs scrolling on the display screen. Specifically, setting the horizontal register for R10 (E80014H) and the vertical register for R11 (E80016H) in terms of dot units will perform scrolling in the corresponding location. (2) Simultaneous/Single Access In light pen mode, enables simultaneous access to each plane of the text VRAM, or single access mode to perform single access with multiple simultaneous access. Specifically, in the case of single access mode, setting R21 (E8002AH) D08 allows simultaneous/single access to multiple planes of text VRAM for each address on D07-D04, respectively. If simultaneous access mode is chosen, please note that the specified planes will be accessed simultaneously. (3) Raster Copy Any raster (raster unit = 4 raster units) can have its address (0-255) copied to another raster position as a 4 raster line segment (1024 raster lines in each direction - up and down). Specifically, setting the source address of the raster copy registers R21 (E8002AH) D03-D00 to define the plane to copy from, then setting the source address for the destination raster registers R22 (E80020CH) D15-D08 and the destination address for the destination raster registers R DO7-D00 as the address to where the raster copy is performed. Therefore, if R21 (E8002AH) D03-D00 is set to '0', raster copy will not be performed. For raster copying that uses the destination video RAM address during the horizontal blank period of the CRT controller source, once the raster line to be copied is allocated to the SAM (raster unit memory), and when the destination video RAM address is set, the horizontal blank period data is used to transfer the memory data to SAM. (4) Bit Mask The text VRAM read can be processed in 16-bit diagonal units in the horizontal direction. Specifically, a mask can be set for each 1 bit in the 16-b diagonal space (0-15) in 128X1 dot units. By setting bit D15-D08 in R23 (E8002EH) and bit mask 7 in D15-D00, then setting bit D09 in R21 (E8002AH) to '1' to use the bit mask. As for the speed in the text screen VRAM, uses simultaneous access and raster copy together. Be careful as using raster copy will have effects beyond mere display and can lead to actual data being affected.

Special CRT Controller Features (Graphics) The Graphics screen has the following features: (1) Scroll Performs scrolling on the display screen. Specifically, setting the names of R12 (E80018H) and R13 (E8001AH) for the actual display 1024x1024 in terms of left and right scroll registers and vertical register in terms of dot units, scrolling is performed at that location. In the case of actual screen size of 512x512, the register (R12-R19) data corresponding to the graphics screen data is set to perform scrolling.

(2) High-Speed Clear

A function to clear the contents of the graphic VRAM at high speed. Specifically, the graphic display plane for high-speed clear of D00 to D02 is selected by R21 (E8002AH), and the graphic VRAM is cleared at high speed using E80480H D01. In this high-speed clear, the rising edge of the VDISP signal of E80480H D01 is latched, and the CRTC writes '0' (that is, clears the data) to the memory in the graphic VRAM every 1 horizontal blanking period by using the SAM (raster unit of the memory) inside the CRTC, resulting in the end of the period (2 vertical periods in the case of interlace).

(3) Image Capture

The function to capture TV and video images by connecting a digital sync tip option and using E80480H D00. That is, this image input is latched on the rising edge of the VDISP signal of E80480H D00 bit and the CRTC writes the image data of 1 screen to the graphic VRAM in the memory SAM (raster unit) inside the CRTC every 1 horizontal blanking period, ending at 1 vertical period (2 vertical periods in the case of interlace). However, this image input will not stop unless '0' is written to E80480H D00.

Page 29

4-3. Sprite This device is equipped with a unique sprite IC, and the sprite screen is independent of the text screen and graphics screen. By using this sprite hardware, you can move any sprite pattern smoothly on a dot-by-dot basis. You can also use it in combination with the text and graphics screen's priority to configure various game screen structures. Table 4-2 shows the specifications of the sprite IC and address map.

Table 4-2 Sprite Specifications

Item	Details	Remarks
Sprite Pattern Definition		
Size	16x16 dot pattern	
Fixed Number	128 patterns in general	BG is not displayed Can define up to max 256 patterns
Colors	16 colors per 1 pattern	16/65536 colors (dot units)
	Total for screen	256 colors/65536 colors
Display		
SP Display		
SP Display High Resolution	1024x1024 dots	
Display Units		
Display Control	Horizontal: 512 dots or 256 dots Vertical: 512 lines or 256 lines	
Display Units	128 SP/screen 32 SP/line	
Additional Features	H-Flip, V-Flip, BG Priority	Priority (BG0>BG1) SP Priority (SP0>SPn>SP127) (These features can be set on a per SP basis.)

| Background Pattern | | | | Display | | | | Size | 8x8 dot pattern | | | | 16x16 dot pattern | | | Fixed Number | 8x8 dot pattern | max 256 patterns | | Colors | Shared with BG pattern and | | | | sprite patterns | | | | 16 colors per 1 pattern | 16/65536 colors (dot units) | | | Total for screen | 256 colors/65536 colors | | Display | | | | BG Display | Text scroll type | max 1024x1024 dots | | Max Screens | Can display max 2 screens | (2 independent scrolls possible) | | Display Units | | | | Display Control | Horizontal: 512 dots or 256 dots | | | | Vertical: 512 lines or 256 lines | | | | When BG1 is displayed with 512 lines | (BG2 pattern size is fixed | | | 16x16 dots | to 16x16 dots for display) | | | 256 dots fixed when displayed with BG2 | BG2 is also displayed in 8x8 dots fixed |

| Additional Features | H-Flip, V-Flip | | | | (These features can be set | | | | on a per BG basis.) | |

4-4. Video Controller This device's video controller has three internal registers, each having the following functions:

● Reg.1 · Setting the screen size of the graphics · Setting the color mode of the graphics memory

● Reg.2 · Setting the priority between graphics, text, and sprites on the screen · Setting the priority of the graphics screen

● Reg.3 · Setting the semi-transparent mode (This function enhances the priority of the graphics screen in a specified area on the display screen.) · Setting the display mode of graphics, text, and sprites

When accessing the registers inside the video controller, please follow the same procedure as when accessing the CRTC signal, ensuring that the V-DISP signal within the MFP0 GPIP4 is set to ”0”.

4-5. Super Impose and Over Scan This device supports the following two display modes:

(1) Low-resolution mode (Horizontal scan frequency 15.98kHz, Vertical scan frequency 60.52Hz) · As a display mode, it supports two modes: Copy screen and Super Imposed

screen. (When connected to a home-use display.) · In Over Scan mode, the physical screen size displayed on the screen may decrease due to over scanning. (The screen size will largely remain the same both horizontally and vertically.) (Interlace)

Physical Screen Size in Low-Resolution Mode	Actual Display Screen Size
256×256	Approximately 236×236
512×512	Approximately 471×471

[Overscan] | [Normal Scan] (Traditional Display Method)

(Note: The dotted line represents the overscanned computer screen.)

- In this mode, in addition to the traditional superimpose modes from the X1 and X1turbo series, interlace-method pseudo high-resolution superimpose modes are also supported. In either case, it's overscan.

```
Traditional Superimpose Mode:
- Pseudo High-Resolution Superimpose Mode
- 256x256
- 512x512 (Interlace)
```

[Traditional Superimpose Mode] [Pseudo High-Resolution Superimpose Mode]

(Note: The dotted line represents the overscanned computer screen.) Access the same memory data in both the odd and even fields of television scan lines. Access different memory data in the odd and even fields of television scan lines.

(2) High-resolution image mode (horizontal sync frequency: 31.5kHz, vertical sync frequency: 55.46Hz)

- As a display mode, the computer screen is supported, but the superimposed screen is not supported.
- The computer screen only displays in normal scan.

[NORMAL SCAN]

CRT screen display area

Border

Computer screen

The 256 x 256 two-tone reading mode is performed in the same way as the X1turbo high resolution 200-line mode. (However, the horizontal and vertical sync frequencies are different.)

5. Miscellaneous Switches (1) This unit is equipped with the following 3 types of switches.

(A) Reset Switch

- When you press this switch, a hardware reset is performed, and the processing begins at the address written in main memory 00000H.

(B) Interrupt Switch

- When you press this switch, the lowest bit of the status register and the NMI (non-maskable interrupt) signal come in, an external interrupt request is made to the address written in main memory 00007CH. The NMI processing routine writes 1 to E8E007H D02 and then performs an NMI reset. If D02 is not written to 1, the NMI processing routine is exited, and the NMI process is invoked again.

(C) POWER Switch

- The POWER switch has two modes: ON and OFF. When on (Vcc1 ON), the circuit operates at one stage. In the OFF mode, MPFP = POWER SW is in ”OFF” (MPFP level 2 interruption), processing before sending MPFP to the MPU. If the MPU does not detect this condition within a control processing routine, it sends a stern instruction ”OFF” ”ON”, ”OFF” to the status register E8E00FH and prevents the missing detection. This prevents access to floppy disks and whenever the power is turned off.

(D) CLOCK SPEED Switch

- By this switch, you can set the clock speed of the MPU to either 10MHz or 16MHz.
- When operating (in Vcc1 ON state) change the clock speed, turn POWER OFF once, then switch and turn the power back on after switching.

Notes:

1) If you switch during operation (Vcc1 ON), the mode does not change.
2) When you perform a reset switch during operation, you can change the mode, but normal operation of the system thereafter is not guaranteed.
3) The reset operation by software cannot change the mode.

Inspection results of power supply systems: In particular, the V1c1 may be turned ON due to the MPF1C failure and can be recognized.

(A) Vcc2 (RTC, SRAM, supply power for various circuits)

- Controlled by the rear power switch, can be turned ON and OFF. Do not use with timers and TV controls. Turn OFF the rear power switch after use.

(B) Vcc1 (RTC, SRAM, EXCLUDING key switches, supply power for almost all other circuits)

- The POWER2 switch is subject to an ON and OFF setting in reverse order of the rear power switch.
- The rear power switch being ON sets the front POWER switch to ON.
- The rear power switch being ON enables the expansion EXPWON signal in the expansion I/O slot.
- The rear power switch being ON enables the RTC2/ALARM timer signal in RTC ALARM time. However, regarding Vcc1OFF, software system board E8E00FH having the value of ”00”, ”OFF” and ”OP” are meaningless.

(C) Backup power supply

- The rear power switch being OFF backs up the RTC SRAM. The backup power supply used to use two manganese dioxide LR batteries, but now it is using primary lithium batteries. Do not operate while charging the batteries.

(3) The LEDs on the front panel light up as follows (assuming Vcc2 is ON).

(A) 16MHz LED and 10MHz LED

- When the front POWER switch is ON and both Vcc1 and Vcc2 are ON, the LEDs correspond according to the MPU operation mode.
- If the POWER2 switch is OFF, it does not correspond to the LED.
- When turning OFF the system with the front POWER switch ON, the LED remains in the state for a few seconds before the 10s internal processing is finished and it turns OFF.
- When Vcc1 is OFF but only Vcc2 is ON, the CLOCK SPEED switching state is shown on the LED as RED.

Vcc2 Vcc1 MPU CLOCK 10MHz LED 16MHz LED operation SPEED SW ON ON 10MHz x GREEN OFF 16MHz x OFF GREEN OFF 10MHz RED OFF OFF 16MHz OFF RED OFF OFF OFF

(B) TIMER LED

- Using RTC and ALARM timer function to set RTC's register address, E8A00H's D041 sets ”1” and turns the TIMER LED on. Conversely, when using the timer function, writing ”0” turns it off. That is, this corresponds to D041, D04 is reset, and when Vcc2 is OFF, it is cut off, and the TIMER LED does not light even though the front POWER switch is ON.

(C) HD BUSY LED

- Indicating the internal hard disk access. It lights up during read/write operations of the hard disk.

Page - 34 -

(D) Floppy Disk Drive LED

Front	Activity LED
POWER Switch ON State	- When media is in the FDDGreen lights up- When there is no media in the FDD and the LED flashing fun

| OFF State | - When there is media in the FDDGreen lights up- When there is no media in the FDDTurns off | - When there is media in the FDDGreen lights up- When there is no media in the FDDTurns off |

(4) This machine uses ports A and B of the 8255 as input ports for two joysticks, and also uses port C as a switching port for controlling the two joysticks, controlling the sound synthesis output and sampling rate switching. Therefore, by using the control word register of this 8255, it is necessary to specify port A and B as input and port C as output.
<8255 Control Word Register (E9A007H)>

D07	D06	D05	D04	D03	D02	D01	D00
1	0	0	0	0	0	0	0

```
0 - Port C (lower 4 bits) output (sound synthesis)
1 - Port B input (joystick No.2)
0 - Port C (upper 4 bits) output (joystick control)
1 - Port A input (joystick No.1)
```

Page 35

6. Keyboard and Mouse

This device uses the keyboard and mouse in the following block configuration.

CPU System Board (EBE007H) Data Bus Control Bus Address Bus

MFP RR SO SI GND TV Controller Remote Signal Dedicated Disk Display

80C51 Vcc2 GND TO RxD TxD T1 80C51 Send Data> <80C51->MFP Receive Data>

GND Vcc2

(CPU System Board (EBE003H))

Notes: Regarding the control of the mouse and TV control, please use either control on the main unit or control on the keyboard side, or use them independently.

Figure 6-1 Keyboard and Mouse Surrounding Block Diagram

(1) Transformation and transmission of key scan data to MFPA (refer to Table 1-3) Unlike the X18&X1turbo series, this keyboard transmits the scanned key data to the sub-CPU without converting it to ASCII code. The serial data of one byte is then sent with information on whether the key was pressed or released. Regarding repeat functionality, this keyboard has the capability to control the start time (Delay 1) and the repeat interval (Speed), as well as the functionality to send the data of the pressed key (MSB) after the delay. This repeat functionality is valid for all keys (refer to Table 6-2).

(2) Transmission of TV control remote control signals In the X18&X1turbo series, the TV control remote control signals are output to P27 (pin 38) of the main unit's 80C49 or P27 through 80C51 (port 1 TI). However, this keyboard has two ways to transmit these TV remote control signals from the main unit's MFPA: via software method and 1-byte data transfer from 80C51. Especially for this keyboard's own TV remote control mode, sending the 1-byte data to MFPA through 80C51 validates the data as a TV control remote command sent from the main unit. Additionally, other system mode management is handled by the system board E8E003HJ D03. By this, the TV control remote signal can be sent through software timers. The commands valid for repeat control are TV Volume Up, Volume Down, Channel Up, and Channel Down.

(3) LED Lighting Control on the Keyboard The illumination of the seven LEDs on the keyboard can be controlled via software. In other words, it's possible to turn on, off, or flash arbitrary LEDs based on one-byte data transmitted from MFPA to 80C51. Moreover, LEDs are lit up when a specific key code generated by the keyboard reset signal is received; thus, it is possible to ascertain that the keyboard has been initialized by the main unit. Additionally, when the keyboard is reset (e.g., during power-up reset or keyboard replacement), the key code transmitted to the main unit indicates the LED status which is used to inform that the keyboard has been reset.

(4) Mask Control Signal (MSCTRL) Control Transmitting one-byte data from the keyboard's MFPA to 80C51 allows control of the MSCTRL signal. This is significant when the keyboard's mouse connector is used.

(5) Key Scan Data Transmission Error Code 80C51 outputs scan data directly via main and interrupt loops. However, if certain error prevention mechanisms within the MFPA or during the RR signal for data reading are unsuitable (triggering issues such as the system board E8E007HJ D01 becoming inactive), then key data transmission by 80C51 is invalid. In such a case, the keyboard enters an idle state until the data becomes acceptable for system board E8E007HJ D01 acknowledgment. This error state is resolved when the RDY signal transitions from a ”0” state, in which 1-byte data transmission is permissible.

Table 6-1 Key Data Transmission Format

Baud Rate: 2400 baud

Key Data Transmission Sequence (Serial Transmission Asynchronous Communication)

| Start Bit | 1 | | Data Length | 8 | | Stop Bit | 1 | | Parity None | |

(6) Key Assignment for X1 Control in Designated TV Control Keys

For the following TV control keys, sending 1-byte data from MFP to 80C51 allows them to be used as TV control keys for X1 punch.

Operation Key	CZ-600C	X1 Punch Mode
TV Control Key ++	Superimpose/Superimpose Toggle Key	Superimpose
TV Control Key +=	TV/External Input Toggle Key	TV
TV Control Key +	TV/Computer Screen Toggle Key	Computer Email

(7) Control of TV Control Mode and Mouse Control Mode

By sending 1-byte data from this MFP to the 80C51, you can toggle the ON/OFF state of the keyboard's TV control key or mouse control key, respectively. Typically, like the SHIFT key, if you hold down a TV control key or mouse control key while pressing another key, the key will be controlled. In this control mode, these toggle operations retain their state until pressed again for the second time, which cancels the previous key state. In this mode, once the TV control key or mouse control key turns ON, it will remain ON until turned OFF, and it keeps holding the key down functionality. Standard key codes are sent from the MFP.

Table 6-2 Key Scan Data Format:

D7 D6 D5 D4 D3 D2 D1 D0 | Key Scan Code |

```
0 - Key pressed state
1 - Key not pressed state
```

1-byte data from 80C51 to MFP Key code reference is in figure 3-2

CZ-634C-TN CZ-644C-TN

7. Sound Function

This machine uses the YM2151 as the FM sound source LSI and the MSM6258 as the voice synthesis LSI.

Fig. 7-1 Sound System Block Diagram

Labels in the diagram:

- Data
- Address
- Control
- Buffer
- Decoder
- FM Sound Source (YM2151) Vcc1 Vss CLK(4MHz)
- μPD8255 PC Vcc1 Vss
- ADPCM (MSM6258) Vcc1 Vss CLK(4MHz)
- PCM PAN
- Built-in Speaker
- Headphone Out
- Amp
- L line Out
- R line Out
- R TV AUDIO
- L TV AUDIO
- line In

7-1. FM Tone Generation

● LSI YM2151 (1) Up to 8 tones can be generated. However, for sinusoidal waves, up to a maximum of 32 tones can be generated. (2) Noise generation. (3) Pitch and time modulation. (4) In addition to the fundamental wave, allows irregular modulation of the upper harmonics. (5) Allows detuning of the 7th harmonic. (6) Adjustment of pitch with 1.6 cent accuracy. (7) Vibrato and tremolo operation. (8) Enables various sound effects due to deep vibrato and frequency modulation as well as detuning of the upper harmonics along with the fundamental wave. (9) Includes two types of timers.

7-2. Voice Synthesis

This system uses MSM6258 in ADPCM method as the voice synthesis LSI. PCM's output control, sample frequency conversion, and switching are performed by the 8255's ports and Ck. The LSI converts voice input to digital via S&H (Sampling & Hold) and AD converter, and when using the PCM (Pulse Code Modulation) code, the difference between the predicted value and the actual value is quantized using the ADPCM (Adaptive Differential PCM) method, reducing the amount of information. Additionally, through the above process, the resulting output is converted back to its original form using a digital-to-analog converter (DAC).

(End of page 41)

CZ-634C-TN
CZ-644C-TN

8. Peripheral LSI

8-1. DMAC

In this book, the 63450 is used as the DMAC. This is an independent 4-channel DMAC, and in this machine, it is allocated as shown in Table 8-1.

Table 8-1 Allocation of each DMAC channel

Channel No.	Allocation	Transfer Demand	Block Transfer
0	Internal 2HD	External Transfer Request	Cycle Steal Mode
1	SCSI Device	Auto Request Max Speed	Full Channel Dual Address Mode Full Channel (programmable)
2	Memory Memory	Auto Request Limited Speed and Max Speed	Continue Mode Array Chain Mode
3	Sound Synthesis	External Transfer Request	Cycle Steal Mode Link Array Chain Mode

Figure 8-1 DMAC Block Diagram

- 68000
- Address Bus
- Data Bus
- Control Bus
- CLK (10/16MHz)
- Vcc1
- Vss
- 63450
- CLK (10MHz)
- Vcc1
- Vss

- Ch0 Internal 2HD
- Ch1 SCSI Device
- Ch2 Memory Memory
- Ch3 Sound Synthesis

This DMAC has the following features. Also, Table 8-2 shows the DMAC transfer request methods, and Table 8-3 shows the DMAC data block transfer.

(1) Equipped with 4 independent DMA channels (priority is programmable). (2) Enables transfers between memory to memory, and memory to I/O devices. (3) Supports block transfer functions of Continue Mode, Array Chain Mode, and Link Array Chain Mode. (4) Programmable transfer using internal registers. (5) Supports error detection and high-reliability data transfer functions with vector error splitting and exception handling. (6) Maximum 5 MBytes/sec. (10MHz) (7) Compatible with the 68000 bus.

Table 8-2: DMAC Transfer Request Methods

Transfer Request Methods

- External Transfer Request (using the REQ pin)

```
- Cycle Steal Mode
  - Transfers at each operation's timing. Releases the bus at each transfer completion.

- Cycle Steal Burst Hold Mode
  - Transfers at each operation's timing. However, does not release the bus for a fixed
  ↪  period after transfer.

- Burst Mode
  - Transfers multiple operations consecutively.
```

- Auto Request (not using the REQ pin)

```
- Fixed Speed
  - Transfers multiple operations consecutively. However, periodically releases the bus
  ↪  during certain intervals and does not monopolize the device.

- Maximum Speed
  - Transfers multiple operations consecutively. However, releases the bus during certain
  ↪  intervals and monopolizes the device until the transfer is completed.
```

- Auto Request (only transfers the first operand) + External Transfer Request (transfers operand from the 2nd one onward)

Table 8-3: DMAC Data Block Transfer Methods

- Single Data Block Transfer
  - Assigns the transfer address and transfer word count to internal DMAC registers.
- Multiple Data Block Transfer

```
- Continue Mode
  - Assigns the transfer address and transfer word count to internal DMAC registers, and
  ↪  sets the CNT bit to indicate the presence of the next data block.

- Array Chain Mode
  - Creates an array table in main memory.

- Link Array Chain Mode
  - Creates a link array table in main memory.
```

8-2. Floating-Point Arithmetic Co-Processor

In this device, it is possible to optionally use the floating-point arithmetic co-processor (hereinafter abbreviated as FPU) MC68881 (16.67MHz).

![Diagram of FPU signal block] MC68881 Main Features ● 8 80-bit floating-point data registers ● 67-bit arithmetic device ● 67-bit barrel shifter and high-speed shifter ● 46 instructions ● Complete compliance with the IEEE 754 standard ● Support for functions not defined by the IEEE standard (transcendental functions such as trigonometric and logarithmic functions) ● 7 types of data types (byte, word and longword integer, single-precision real number, double-precision real number, extended-precision real number, packed BCD real number) ● Support for 22 constants in on-chip ROM (including π) ● Virtual memory compatible operation ● Efficient handling of exceptions, interrupts, and inserted processing mechanisms ● Parallel execution of main processor commands ● Compatible with hosts with 8-bit, 16-bit, or 32-bit data buses The IC slots on the FPU-compatible motherboards detect the presence of an FPU or not, and automatically judge whether an FPU is installed in the motherboard IC slot. If the FPU is not installed in the motherboard IC slot, connecting it to the corresponding CPU socket on the CZ-6BP1 expansion slot will enable proper function. However, if the FPU is installed on both the motherboard and the I/O slot processor board, the system will recognize the FPU on the motherboard as the priority and operate accordingly. Therefore, ensure not to have duplicate installations.

8-3 Expansion Main Memory (1) Internal Expansion Board (CZ-6BE2A, CZ-6BE2B)

- Expansion by 2MB units up to a maximum of 6MB (Actual maximum with internal extension of 8MB) Installed on expansion board
- 2MB Memory Install Range Installation range: 200000~3FFFFF H
- Connector for 2MB Memory Module

```
Connector 1: 400000~5FFFFF H
Connector 2: 600000~7FFFFF H
```

- BIOS ROM Mount IC Socket (2 of 1MB EPROM)
- BIOS ROM Replacement Switch
- The memory controller automatically determines the presence or absence of the above three 2MB memory boards by checking the following three signals (Each signal is enabled when LOW level):

```
DRAMSENSE1 Signal: 200000~3FFFFF H Memory
DRAMSENSE2 Signal: 400000~5FFFFF H Memory
DRAMSENSE3 Signal: 600000~7FFFFF H Memory
```

If this function is not used and the internal expansion board is not used, the previously used IO slot 2.4MB memory board attachment is enabled. (CZ-6BE2,4)

- Accessible in sync wait at both 16/10MHz.

(2) Adding memory on IO slot

```
Possible to expand memory according to internal switching range
* In the case of 8MB installed, 4MB can be expanded on the IO slot
* In the case of 6MB installed, 6MB can be expanded on the IO slot
* In the case of 4MB installed, 8MB can be expanded on the IO slot
* In the case of 2MB installed, 10MB can be expanded on the IO slot
```

(Note) IO slot memory boards are inaccessible in non-wait state at 16MHz mode. (Possible at 10MHz mode)

8-4. MFP

In this unit, we use the 68901 from the 68000 family for data transmission and reception with the keyboard, timer functions, and various interrupt controls.

CLK RESET Vcc1 Vss (4MHz)

Internal Control Logic

Timer C, D (Delay Mode)

- XTAL1 (4MHz)
- XTAL2 (4MHz)
- TCO (N C)
- TDO (N C)

Timer A (Event Count Mode)

- TAO (N C)
- TAI (CRT0 V-DISP signal)

Timer B, (Delay Mode)

- TBO (USART serial clock)
- TBI (N C)

USART (Keyboard)

- Rx (Reception Data)
- SI (Keyboard/Reception Data)
- RC (TBO input)
- SO (Keyboard send data)
- TC (TBO input)
- Tx (Send Data)

GPIP

- GPIP0 (RTC Alarm signal based on detection of ON/OFF)
- GPIP1 (EXPWON signal based on detection of ON/OFF)
- GPIP2 (POWER SW signal based on detection of ON/OFF)
- GPIP3(VCC2 drop signal detection)
- GPIP4 (CRT0 V-DISP signal detection)
- GPIP5 (RTC CLKOUT (1Hz) signal)
- GPIP6 (RTC 1QR signal)
- GPIP7 (CRT0 H-SYNC signal)

Interrupt Control

- IRQ
- IEI
- IEO
- IACK

Data (8) Register select (5) CS R/W DS DTACK

CPU Bus I/O

Figure 8-3 MFP Block Diagram

Page 46

<68901MFP Interrupts, Read Address Channel>

Decoder

68000MPU

| High | INT7 IPL0 | Priority | INT6 IPL1 | Priority | INT5 IPL2 | Priority | INT4 | Low | INT3 | | INT2 | | INT1

| Priority bits: High | 15 | GPIP7 (CRTC's H-SYNC signal)

```
           14   | GPIP6 (CPTC's IRQ signal)
           13   | TimerA (CPTC V-DISP signal)
           12   | Receive Buffer Full (Data reception)
           11   | Receive Error (Data reception error)
           10   | Transmit Buffer Empty (Data transmission)
            9   | Transmit Error (Data transmission error)
            8   | TimerB (USART serial clock)
            7   | GPIP5 (RTC 1Hz clock signal)
            6   | GPIP4 (CRTC V-DISP signal monitor)
            5   | TimerC (General 8-bit timer)
            4   | TimerD (General 8-bit timer)
            3   | GPIP3 (Power supply detection)
            2   | GPIP2 (POWER SW ON/OFF detection)
            1   | GPIP1 (Expansion slot detection or EXPWON signal ON/OFF detection)
```

Low | 0 | GPIP0 (RTC ALARM signal ON/OFF detection)

Figure 8-4 MFP interrupt block diagram

(TAI) (CRTC's V-DISP signal)

(4MHz)

Figure 8-5 MFP timer block diagram

CZ-634C-TN CZ-644C-TN

Inside the main diagram:

TBO terminal RC terminal TC terminal

USART (x16)

SI (Keyboard receive data) SO (Keyboard send data) RR (Receive status) TR (Transmit status)

Box on the right side of the diagram:

Asynchronous communication 2400 baud Start 1 bit, Data 8 bits No parity, Stop 1 bit

Fig. 8-6 MFP USART System Block Diagram

Table 8-4 Details of MFP Channels

Channel No.	Function Details
QPIP7	Generates an interrupt when a rising edge of the CRTC H-SYNC signal is detected. (Set ”0” to the a
QPIP6	Generates an interrupt when the programmed edge (R09(E80012H)) of the CRTC is detected, or ge
TimerA	Generates an interrupt when the rising edge of the CRTC V-DISP signal is detected. In Pulse Widt
Receive Buffer Full	Generates an interrupt when data is received by the keyboard (when reception data is transferred
Receive Error	Generates an interrupt when an error occurs during data reception by the keyboard (overrun erro
Transmit Buffer Empty	Generates an interrupt when data is transmitted from the keyboard (when transmission data is tra
Transmit Error	Generates an interrupt when an error occurs during data transmission by the keyboard (unexpec
TimerB	Generates an interrupt when the internal clock (4MHz) is input (delay mode), and changes the M
QPIP5	Generates an interrupt when the state change is detected on the signal of the RTC CLKOUT(1Hz).
QPIP4	Reads the state of the CRTC V-DISP signal. When ”1” and ”2”, it indicates the vertical sync period,
TimerC	Generates an interrupt when the internal clock (4MHz) is input (delay mode), and uses the arbitra
TimerD	Generates an interrupt when the internal clock (4MHz) is input (delay mode), and uses the arbitra

Channel No. / Function Details

GPIP3 When Vcc2 (secondary main power source) is detected, it reads whether it has been reinserted. An ”H” indicates detection of Vcc2, while a ”0” indicates ”L”. When Vcc2 is detected, GPIP3 automatically becomes ”0”. However, if Vcc2 is detected again after being cut off, GPIP3 will change to ”1”. If GPIP3 remains ”0” even after Vcc2 is detected, it signifies that the front power (POWER switch) is ON. When the POWER switch is ON, all subsequent detections will show ”12” on GPIP3 instead of ”0”. The MPU will determine if Vcc2 is inserted and must set GPIP3 to ”1” by writing ”12” to E8A0 and D056 registers. Access the RTC registers with the previously saved values. Generally, reading will prevent using the port report.

GPIP2
The POWER switch reads whether the computer power (Vcc1) is ON or OFF. Normally, the switch state is ”0” or ”L” (OFF), but pressing it changes it to ”1” or ”H”. When pressing the switch again, it reverts. Ensure the signal step prevents the malfunction.

GPIP1
Reads whether the computer power (Vcc1) is ON by using the EXPWON signal from the expansion bus slot. When external sources except EXPWON are given, it is ”0” or ”L”. Verify if the computer power is ON by checking GPIP1.

GPIPO
Uses RTC ALARM timer to generate ALARM signals, and determines if the computer's power (Vcc1) is ON with the ALARM signal. ”H” on power ON remains ”H” without ALARM signals. After a minute, it goes ”L”. Monitoring GPIP0 within 1 minute reveals if power was ON or OFF. This signal, combinable with 1Hz and 16Hz clock pulses, can generate any status changes based on the signal power.

Additionally, regarding GPIP0 to GPIP3, it checks from where the computer power was turned ON. The flowchart is shown below.

[Power ON/OFF]

When Vcc1 turns ON Check MFP's GPIP2

```
                "0"                       "1"
                   ↓                         ↓
```

Power switch is turned ON Check MFP's GPIP1

```
                     "0"                             "1"
                   ↓                                  ↓
```

Turned ON by the EXPWON signal from the expansion I/O slot Check MFP's GPIP0

```
                     "0"                                                  "1"
                   ↓                                                        ↓
```

Turned ON by the ALARM signal from the RTC ALARM timer Recheck GPIP2, GPIP1, and GPIP0. After final verification, write “00H” to system port E8E00FH to “OFF” and turn off power

The page number at the bottom reads ”51.” The model numbers mentioned at the top right are CZ-634C-TN and CZ-644C-TN.

Top Left: CZ-634C-TN CZ-644C-TN

Middle Right: RS-232C Mouse Connector Pinout

Bottom Center: Fig. 8-7 SCC Block Diagram

Bottom Center:

Other Labels in the Diagram: TxDA, RTSA, DTRA, D/C, A/B, RxDA, CTSA, DCDA, RxCA, CTCA, CTCS, DCDB, DTRB, CE, RD, WR, DO-D7, PCLK, TRxCA, RTSB, RxDB, IRQ, INT, INT ACK, IEI, GND, Vcc

(5MHz) Vcc2, 4.7 kΩ, Vcc1, GND RTC, CTS, DSR, ST1, ST2, CI, CD, SG, FG Mouse MSCCTRL, MSDATA, GND

75188 +12V, -12V, Vcc, TxD, RTS, DTR, ST1

75189A (290pF × 4), RxD, CTS, DSR, RT

75189A (290pF × 3), ST2, CI, CD, SG, FG

Keyboard MSDATA

8-5. SCC

In this manual, the RS-232C serial communication controller is used to support the mouse. We use the Z8530 SCC for the Z8000 family. The block diagram is shown in Fig. 8-7.

(1) Channel A (RS-232C)

```
* Asynchronous Mode
  - 5, 6, 7, 8-bit character
  - 1, 1.5, 2 stop bits per character
  - Even parity, odd parity, no parity
  - x1, x16, x32, x64 clock rate
  - Block generation and detection
  - Parity, overrun, and framing error detection

* Synchronous Mode
  - Byte-oriented synchronous mode: The character synchronization can be internal, external, or either
    1 to 2-character synchronization
    The synchronization character can be 6 to 8 bits
    The automatic insertion or exclusion of synchronization characters
    CRC generation and verification
  - SDLC/HDLC mode: Generation and detection of address field
    Automatic zero insertion and deletion
    Automatic flag insertion between messages
    Detection of address field
    Information field protection
    CRC generation and verification
    EOP detection and escape (online escape) in SDLC loop mode

* Data Transfer Rate
  - Maximum 1.5 Mbits/sec (NRZ signal mode)
  - Maximum 375 Kbits/sec (FM encoding mode using DPLL)
  - Maximum 187 Kbits/sec (NRZI encoding mode using DPLL)
```

(2) Channel B (Mouse)

```
* Asynchronous Communication
  - Baud Rate: 4800 baud
  - Start Bit: 1 bit
  - Data Bits: 8 bits
  - Parity: None
  - Stop Bits: 2 bits
  - Data Bus: RXDB
  - Control Bus: RTSB
```

(3) SCC Register Details

For the access to each write/read register (write-only register WR3 and read/write register RR8 are excluded), “Command Mode” must be utilized. First, select the register to access using bits D00- D02 of WR0. After designating the accessed register, reading/writing to the data register will allow you to read/write data. Here, access to each register is configured through the reading/writing of the data.

Also, concerning the access to write-only register WR8 and read/write register RR8, you can read/write data by accessing the Data Port. Table 8-5 shows the baud rate generator clock number for each baud rate.

Table 8-5 Clock number for each baud rate

Baud Rate	Clock for 5MHz (in hexadecimal)	Remarks
9600	(14(000EH))	5×10^6
4800	(31(001FH))	Clock number = ---------
2400	(63(003FH))	2
1200	(128(0080H))	2 × (Baud rate × 16)
600	(258(0102H))	
300	(519(0207H))	However, when the input clock (PCLK) is 5MHz
150	(1040(0410H))	and data speed uses a 16x clock.
75	(2081(0821H))	

8-6. RTC This machine uses RP5C15 as the Real-Time Clock (RTC).

RTC Block Diagram

+---------------------------+ | RP5C15 | | | | +-------------------------+ | | | Address (4-bit) | | | | Data (4-bit) | | | | Control | | | | Battery Backup/Vcc2 | | | +-------------------------+ | | | | | +---------------+ | | |CKOUT (1Hz) Clock | | +---------------+ | | | | | ALARM (Alarm signal, 1Hz, | | 16Hz Clock) | | | | +------------|--------------+

| | | | | +------------------------+ | | GPI(PF) | | | Write/Read | | +------------------------+ | | +------- POWER LED & TIMER LED ON/OFF

| | | | | +------------------------+ | | GPI(PF) | | | Read | | +------------------------+ | | +--------+

This RP5C15 has the following features with the same READ/WRITE sequence of the memory providing the settings and reading outputs of time using real-time clock.

(1) Possible to connect directly to the 16-bit CPU and allows high-speed access (2) Address and data bus (D00-D03) (3) 4-bit address input (4) Embedded counter (year, month, day, hour, minute, second) calendar (supports until the year 2099) (5) Internal alarm function (6) All time data can be displayed in BCD code (7) Internal 30 seconds adjust function (8) Battery backup possibility (requires 2.0V) (9) The alarm signal or 1Hz timing pulse can be output as 16Hz

Figure 8-9 RTC Block Diagram

Page -54-

(10) 1Hz Clock Output In particular, this device uses the RP5C15’s alarm feature and clock output as follows.

- Alarm Feature (Alarm Interrupt) It is used for setting the scheduled timer for turning on the computer's power (Vcc1 ON). When the set date and time match the current date and time, the power to the computer is turned on and the ALARM signal is output, triggering the alarm interrupt. Furthermore, the 1Hz clock output is sent to

MFPO/GPIO ● input (used to divide MPFPI). Specifically, the output from ALARM is software selectable and can be executed by capturing the software interrupt signal to divide the MPFP. Also, when setting the alarm timer, the TIMER LED lights up (write “1H” in E8A0H's D04=1*), additionally, after writing “31H” to the system port EB800DH and enabling WRITE ENABLE, write to SRAM, then turn on the TIMER LED and finally write “0D04=1H” to E8A0H. Set the system port EB800DH to OFF. Be sure to write SRW=READ ONLY=K, otherwise processing will be needed. (For MPFPK, re-process when Vcc2 is re-inserted).

- 1Hz Clock Output The 1Hz clock is output from the CLK OUT terminal and connected to the MFPO/GPIO P5 (used to divide the 7-level interrupt input). By doing this, MPFPK division can be performed. This 1Hz signal is also used to turn on the POWER LED and the TIMER LED on the front panel. Normally, the power backup for the RP5C15 internal power is connected to Vcc2, so please make sure the power switch is absolutely OFF when using timing functions.

9. Peripheral 9-1. Disk The device is equipped with two data-sided FDD (floppy disk drives). This FDD is controlled by FDC and μPD72065 is used. For SCSI interface machines, 80MBHDD is controlled by SCSI controller MB89352. Please refer to figure 9-1.

Table 9-1 Floppy Disk Specifications

| 128, 256, 512, 1,024Bytes/sector | FDCA Input Clock | Formatting Method | | 26, 16, 8 Sectors/track | 2HD(8MHz) | MFM | | 77Track/side | 2DD-2D(4MHz) | FM | | 154Track/disk | | Switchable | | (When formatted: 1.28MBytes) | | |

CZ-634C-TN CZ-644C-TN

Front POWER SW	Activity LED
ON State	- Green light is on when media is in FDD - Green light blinks when media is not in FDD, and LED blink
OFF State	- Green light is on when media is in FDD - Off when media is not in FDD

3. Eject SW Mask Function Eject function masking feature to prevent ejection during access

4. Media Insert/Eject Detection Feature Function to constantly monitor insertion and ejection of each drive

5. Media Mis-insertion Detection Feature Function to detect mis-insertion of media for each drive

6. Interruption Feature Function for interruption during media insertion or ejection

Address Bus Data Bus SCSI Controller Hard Disk Control Hard Disk Data Bus

HD (Hard Disk) FD (Floppy Disk)

Address Bus Data Bus I/O Controller FDD Option Control

OPTION SELECT 0~3 EJECT EJECT MASK LED BLINK FDD INT ERR DISK DISK IN TRKO6 WRITE PROTECT DRIVE SELECT 0~3

FDC (μPD72065) READY INDEX DISK TYPE SELECT DIRECTION STEP WRITE GATE WRITE DATA SIDE SELECT

2HD/2DD-2D Switch Vcc1 Vss 16MHz

VFO Circuit (SED9420AC) MSN/STD MOTOR ON READ DATA

Figure 9-1 FDD/HDD Peripheral Block Diagram

The labels next to specific arrows relate to signals or commands connecting different components in a hard disk and floppy disk drive control system.

Table 9-2 Internal FDD Specifications

Specification	Details
Memory Capacity (KBytes)	Unformatted: 1667 Formatted: 1065 Track capacity: 10.42 IBM standard 26 se
Data Transfer Rate (Kbits/sec)	500
Access Time (msec)	Track Read Time: 3 Seek and Settling Time: 15 Average Access Time: 95 Seek
Media Rotation Speed (rpm)	360
Spindle Motor Startup Time (sec)	0.5
Maximum Internal Recording Density (BPI)	9870
Number of Tracks	TRACK/SIDE: 77 TRACK/DISK: 154
Track Density (TPI)	96
Number of Heads	2
Encoding Method	MFM FM method possible
External Dimensions (mm)	27.0 (H) × 148.0 (W) × 198.0 (D)
Weight	900g
Others	Auto Clamp Eject Mechanism, Auto Carriage Return Mechanism, VFO (SED9

Table 9-3 Internal HDD Specifications

Memory Capacity Formatted: 85,800 (kBytes) Track Capacity: Inner: 20,992 / Outer: 27,648

Data Transfer Speed (Mbits/sec): Inner: 10 / Outer: 13

Access Time (msec): Average Access Time: 19

Disk Rotation Speed (rpm): 2,975 ±1%

Recording Density (BPI): 31,500

Number of Tracks:

```
  TRACK/SIDE: 868
  TRACK/DRIVE: 3,472
```

Track Density (TPI): 1,300

Number of Heads:

Recording Method:

```
  RLL2-7

Other:
  Built-in Disk Controller
  Interface = SCSI
```

(Page 59)

<CZ-634C-TN CZ-634C-TN

Table 9-4 Comparison of µPD72065 and M88877A

µPD72065 M88877A -checks for Read/Write execution result (status) -checks for Read/Write execution result (status) are performed by polling. are performed by polling. -requires I/O ports for driver select, head select -requires I/O ports for driver select, head select signal output. signal output. -each drive's current track position is maintained -each drive's current track position is maintained

```
   within the FD, making program handling easy.                    in the main RAM,
    ↪  necessitating allocation
                                                                  of track numbers, making
                                                                       ↪  program
                                                                       ↪  handling
                                                                    slightly more complex.
```

-Formatting is possible simply by giving -For formatting, data for one track (6KB~8KB)

```
   a byte of data command to the FDC.                           is needed, making program
    ↪  handling slightly
                                                                        more complex.
```

-Depending on the type of error, command -Depending on the type of error, it may not be reset,

```
  may not reset correctly. (e.g. EX) With                          e.g. EX) may not return
   ↪  from hang-up while reading,
  hang-ups in media, the drive may not return                  pulling out the media, etc.
  to normal state if ejected.
```

9-2. Printer This machine, like the X1&X1turbo series, is equipped with an 8-bit parallel printer I/F according to Centronics standard.

To send a byte of data to the printer from this machine, control signals and BUSY signals use STROBE signals. The BUSY signal is active low, preventing data transmission when the level is "H". That is, it waits until the BUSY signal goes "L" for transmission. Also, the STROBE signal is the output signal to the printer. The printer samples data on the rising edge of this STROBE signal. Therefore, be careful when issuing the STROBE signal as before issuing, it is not possible to guarantee the data will be latched in the ECO011 Data Latch Register.

This machine can accept BUSY signal interrupt requests, allowing software masks. Using this function avoids the need to request BUSY signal checks, knowing printer status through interrupts.

Table 9-5 Printer Register Map

Register Address	D07	D06	D05	D04	D03
E8C001H	←-------- Printer Output Data --------→	Printer Data Register (Write Operation Only)			
E8C003H	×	×	×	×	×
E9C**1H	×	×	×	×	(Wr

```
   | | | | | 0 - Printer Busy Interrupt Mask
   | | | | | 1 - Printer Busy Interrupt Mask Disable
   | | | | | 0 - FDD Interrupt Mask
   | | | | | 1 - FDD Interrupt Mask Disable
   | | | | | 0 - FDC Interrupt Mask
   | | | | | 1 - FDC Interrupt Mask Disable
```

E9C**1H | | | | | (Read Operation Only)

```
   | | | | | 0 - Printer Busy Interrupt Mask Enabled
   | | | | | 1 - Printer Busy Interrupt Mask Disabled
   | | | | | 0 - FDD Interrupt Mask Enabled
   | | | | | 1 - FDD Interrupt Mask Disabled
   | | | | | 0 - FDC Interrupt Mask Enabled
   | | | | | 1 - FDC Interrupt Mask Disabled

   | | | | → 0 - Printer Busy Interrupt ON (BUSY=0) - Printer Data Transmission Permitted
  | | | | → 1 - Printer Busy Interrupt OFF (BUSY=1) - Printer Data Transmission Not Permitted
   | | | | → 0 - FDD Interrupt ON
   | | | | → 1 - FDD Interrupt OFF
```

```
| | | | → 0 - FDC Interrupt ON
| | | | → 1 - FDC Interrupt OFF
```

E9C**3H | Interrupt Vector | → | 1 | 1 | Printer Busy Interrupt Vector

9-3. Joystick

```
On X1, X1turbo series, PSG register 14, 15 were used to access two joysticks, but this
↪  machine uses the 8255 port and can use two joysticks. Note, if using two joystick
↪  inputs, set the upper 4 bits (D04-D07) of E9A005H to "0" as shown below:

          mPD8255
```

Address -------> portA ------------> Joystick No.1

```
portB ------------> Joystick No.2
portC ------------> Sound Synthesis PCM Control
```

Figure 9-2 Joystick Block Diagram

Register Address	D7 D6 D5 D4 D3 D2 D1 D0
E9A001H	* <--- Joystick No.1
E9A003H	* <--- Joystick No.2
E9A005H	IOC7 IOC6 IOC5 IOC4 Sampling Rate PCM PAN

Table 9-6 Joystick Register Address Map

(1) Set 8255 to mode 92H to specify E9A007H B921. (2) Set the joystick standard mode A to specify all 0 with E9A005H (Port C) higher 4 bits (D04~D07) as 0. (3) Read Joystick Input Data No. 1 from E9A001H. Also, if reading Joystick Input Data No. 2, read from E9A003H.

9-4 Expansion I/O Slot

(1) Most of the signals that appear on the expansion I/O slot are signals of the same timing when the MPU operates in 16MHz mode or in 10MHz mode.

```
- This makes it possible to use peripheral boards for the X68000 even in 16MHz mode.
- However, depending on the board, subtle timing differences in the hardware and software
↪  might cause operational issues, so caution is necessary.
```

(2) If the device on the expansion I/O slot becomes bus master and accesses the system's devices, all signals given by the board to the system must be in the same timing as at the time of 10MHz mode.

Extension I/O Socket Terminal (A)

Pin No.	Signal Name	I/O	Remarks
1	GND		Ground
2	20M	Out	20MHz Clock
3	GND		Ground
4	DB0	I/O	Data
5	DB1	”	”
6	DB2	”	”
7	DB3	”	”
8	DB4	”	”
9	DB5	”	”
10	DB6	”	”
11	GND		Ground
12	DB7	I/O	Data
13	DB8	”	”

Pin No.	Signal Name	I/O	Remarks
14	DB9	”	”
15	DB10	”	”
16	DB11	”	”
17	DB12	”	”
18	DB13	”	”
19	DB14	”	”
20	DB15	”	”
21	GND		Ground
22	+12V		
23	+12V		
24	FC0	I/O	Function Code (Indicates the state of MPU execution)
25	FC1	”	”
26	FC2	”	”
27	AS	”	Indicates valid data on address bus
28	LDS	”	Lower data strobe
29	UDS	”	Upper data strobe
30	R/W	”	Indicates the direction of data transfer based on MPU
31	-12V		
32	-12V		
33	VMA	I/O	Indicates valid data on address bus
34	EXVPA	In	Indicates address is verified by surrounding 68000 family peripheral devices
35	DTACK	I/O	Data transfer acknowledgment complete
36	EXRESET	Out	External reset
37	HALT	I/O	In: MPU halt request, Out: System halt
38	EXBERR	I/O	Indicates abnormality on external bus operation
39	EXPWON	In	External bus on
40	GND		Ground
41	Vcc2	+5V	
42	Vcc2	+5V	
43	Vcc2	+5V	
44	SELEN	Out	Main memory address row/column switch signal
45	CASRDEN	Out	Main memory CAS signal (during read)
46	CASWRL	Out	Main memory CAS signal (lower data when light)
47	CASWRU	Out	Main memory CAS signal (upper data)
48	INH2	Out	Main memory refresh cycle
49	Vcc1	+5V	
50	Vcc1	+5V	

(Page - 64 -)

Terminal No.	Symbol Name	I/O	Remarks
1	GND		Ground
2	10M	Out	10 MHz clock
3	10M	Out	10 MHz clock inverted signal
4	E	Out	Enable (68000 series peripheral control)
5	AB1	I/O	Address
6	AB2	”	”
7	AB3	”	”
8	AB4	”	”
9	AB5	”	”
10	AB6	”	”
11	GND		Ground
12	AB7	I/O	Address
13	AB8	”	”
14	AB9	”	”
15	AB10	”	”
16	AB11	”	”
17	AB12	”	”
18	AB13	”	”
19	AB14	”	”

Terminal No.	Symbol Name	I/O	Remarks
20	AB15	”	”
21	GND		Ground
22	AB16	I/O	Address
23	AB17	”	”
24	AB18	”	”
25	AB19	”	”
26	AB20	”	”
27	AB21	”	”
28	AB22	”	”
29	AB23	”	”
30	IDDIR	Out	Data bus transection direction control signal
31	-		-
32	HSYNC	Out	Horizontal sync signal
33	VSYNC	Out	Vertical sync signal
34	DONE	I/O	Block data transfer complete (DMA)
35	DTC	Out	Device data transfer complete (DMA)
36	EXREQ	In	External request (DMA)
37	EXACK	Out	External permission (DMA)
38	EXPCL	I/O	External peripheral control (DMA)
39	EXOWN	I/O	External OWN (DMA)
40	EXNMI	In	External NMI
41	GND		Ground
42	IRQ2-n	In	Interrupt request (n: slot 1 or 2)
43	IRQ4-n	In	”
44	IACK2-n	Out	Interrupt acknowledge (n: slot 1 or 2)
45	IRCK4-n	Out	”
46	BRn	Out	Bus request (n: slot 1 or 2)
47	BGn	Out	Bus grant (n: slot 1 or 2)
48	BGACK	I/O	Bus grant acknowledge (indicates other devices have bus master)
49	Vcc1		+5V
50	Vcc1		+5V

This table appears to reference the pin designations and functions for an expansion I/O slot.

CZ-634C-TN CZ-644C-TN

9-5. Various Connectors (The pin numbers of the connectors represent the view from the part side.)

◎ Analog RGB Signal Output Connector

Pin No.	Signal Name	I/O	Remarks
1	R OUT	Out	Analog 0.7Vp-p (75Ω termination)
2	GND	Out	Ground
3	G OUT	Out	Analog 0.7Vp-p (75Ω termination)
4	GND	Out	Ground
5	B OUT	Out	Analog 0.7Vp-p (75Ω termination)
6	GND	Out	Ground
7	YS	Out	Indicates presence of computer data
8	GND	Out	Ground
9	N.C	-	Not connected
10	AUDIO L	Out	Audio signal left
11	AUDIO R	Out	Audio signal right
12	GND	Out	Ground
13	N.C	-	Not connected
14	HSYNC	Out	Horizontal sync signal TTL level
15	VSYNC	Out	Vertical sync signal TTL level

◎ TV Control Connector

Pin No.	Signal Name	I/O	Remarks
1	EXHSYNC	In	External horizontal sync signal TTL level
2	EXVSYNC	In	External vertical sync signal TTL level
3	TVPOWERON/OFF	Out	TV power on/off signal
4	TVREMOTE	Out	TV remote signal
5	Vcc1	Out	+5V
6	GND	Out	Ground
7	GND	Out	Ground
8	N.C	-	Not connected

◎ IMAGE IN (Image Input Connector)

Pin No.	Signal Name	I/O	Remarks
1	ADD11	In	Analog/digital conversion data
2	ADD10	In	”
3	ADD9	In	”
4	ADD8	In	”
5	ADD7	In	”
6	ADD6	In	”
7	ADD5	In	”
8	ADD4	In	”
9	ADD3	In	”
10	ADD2	In	”
11	ADD1	In	”
12	ADD0	In	”
13	Q A	Out	Dot clock
14	Vcc1	Out	+5V
15	GND	--	Ground

(Page - 66)

Terminal No.	Signal Name	I/O	Remarks	Notes
16	Vcc3	Out	+12V	
17	CD4	Out	Computer control signal	
18	CD3	Out	Computer control signal	
19	CD2	Out	Computer control signal	
20	CD1	Out	Computer control signal	
21	CD1	Out	Computer control signal	
22	ADD15	In	Analog/Digital conversion data	
23	ADD14	In	Analog/Digital conversion data	
24	ADD13	In	Analog/Digital conversion data	
25	ADD12	In	Analog/Digital conversion data	

Printer Connector (Pins diagram)

Terminal No.	Signal Name	I/O	Remarks	Notes
1	STROBE	Out	Negative polarity print output strobe signal	
2	PA0	Out	Parallel data bus	
3	PA1	"	"	
4	PA2	"	"	
5	PA3	"	"	
6	PA4	"	"	
7	PA5	"	"	
8	PA6	"	"	
9	PA7	"	"	
10	N.C	-	Not connected	
11	BUSY	In	Printer 'LOW' level when ready	
12	N.C	-	Not connected	
13	GND	Out	Ground	
14	GND	Out	Ground	

SEE THROUGH COLOR (Pins diagram)

Terminal No. | Signal Name | I/O | Remarks | Notes

```
1 | VHT | Out | Video Half Tone (semi-transparent color)
2 | GND | Out | Ground
```

LINE IN (Pins diagram)

Terminal No. | Signal Name | I/O | Remarks | Notes

```
1 | GND | Out | Ground
2 | LINEIN | In | Audio synthesis input
3 | N.C | - | Not connected
```

LINE OUT (Pins diagram)

Terminal No. | Signal Name | I/O | Remarks | Notes

```
1 | GND | Out | Ground
2 | L | Out | Audio (left output)
3 | R | Out | Audio (right output)
```

Joystick Connector Joystick 1

Pin No.	Signal Name	I/O	Remarks
1	IOA0	In	8255 PA0 Pin
2	IOA1	In	PA1
3	IOA2	In	PA2
4	IOA3	In	PA3
5	Vcc1	Out	+5V
6	IOA5	I/O	PA5/PC6
7	IOA6	I/O	PA6/PC7
8	IOC4	Out	PC4
9	GND	Out	Ground

Joystick 2

Pin No.	Signal Name	I/O	Remarks
1	IOB0	In	8255 PB0 Pin
2	IOB1	In	PB1
3	IOB2	In	PB2
4	IOB3	In	PB3
5	Vcc1	Out	+5V
6	IOB5	In	PB5
7	IOB6	In	PB6
8	IOC5	Out	PC5
9	GND	Out	Ground

RS-232C Connector

Pin No.	Signal Name	I/O	Remarks	Pin No.	Signal Name	I/O	Remarks
1	FG	I/O	Protective Earth	14	N.C	-	Not Connected
2	TXD	Out	Transmit Data	15	ST2	In	Transmit Signal Element Timing
3	RXD	In	Receive Data	16	N.C	-	Not Connected
4	RTS	Out	Request to Send	17	RT	In	Receive Signal Element Timing
5	CTS	In	Clear to Send	18	N.C	-	Not Connected
6	DSR	In	Data Set Ready	19	N.C	-	Not Connected
7	SG	I/O	Signal Ground	20	DTR	Out	Data Terminal Ready
8	CD	In	Carrier Detect	21	N.C	-	Not Connected

Pin No.	Signal Name	I/O	Remarks	Pin No.	Signal Name	I/O	Remarks
9	N.C	-	Not Connected	22	CI	In	Ring Indicator
10	N.C	-	Not Connected	23	N.C	-	Not Connected
11	N.C	-	Not Connected	24	ST1	Out	Transmit Signal Element Timing
12	N.C	-	Not Connected	25	N.C	-	Not Connected
13	N.C	-	Not Connected				

(Note: ”-” denotes that the pin is not used or not connected.)

Key Jack

Pin No.	Signal Name	I/O	Remarks
1	Vcc2	Out	+5V
2	MOUSE DATA	In	Mouse Data
3	KEYT × D	In	Key Receive Data
4	KEYR × D	Out	Key Send Data
5	READY	Out	Key Send Permission/Rejection
6	REMOTE	In	Remote Signal
7	GND	Out	Ground

Headphone

Pin No.	Signal Name	I/O	Remarks
1	GND	Out	Ground
2	L	Out	Audio Signal (Left)
3	R	Out	Audio Signal (Right)

Mouse Port

Pin No.	Signal Name	I/O	Remarks
1	Vcc1	Out	Ground
2	MS CTRL	Out	Control Signal
3	MS DATA IN	In	Mouse Data
4	GND	Out	Ground
5	GND	Out	Ground

3D Terminal External Floppy Disk Connection Connector

Pin No.	Signal Name	I/O	Remarks
1	DISK TYPE SELECT	Out	Disk Type Selection Signal
2	N.C	-	Not Connected
3	DRIVE SELECT 3	Out	Drive Select Signal 3
4	INDEX	In	Disk Index Signal
5	DRIVE SELECT 0	Out	Drive Select Signal 0
6	DRIVE SELECT 1	Out	Drive Select Signal 1
7	DRIVE SELECT 2	Out	Drive Select Signal 2
8	MOTOR ON	Out	Motor On Signal
9	DIRECTION	Out	Head Move Direction Signal
10	STEP	Out	Head Move Signal
11	WRITE DATA	Out	Write Data Signal

(CZ-634C-TN / CZ-644C-TN)

Pin No.	Signal Name	I/O	Notes
12	WRITE GATE	Out	Write Gate Signal
13	TRACK 00	In	Track 0
14	WRITE PROTECT	In	Write Protect Signal
15	READ DATA	In	Read Data Signal
16	SIDE SELECT	Out	Head Select Signal
17	READY	In	Drive Ready Signal
18	N.C	-	Not Connected
19	N.C	-	Not Connected
20	OPTION SELECT 0	Out	Option Select 0
21	OPTION SELECT 1	Out	Option Select 1
22	OPTION SELECT 2	Out	Option Select 2 CZ-600C dedicated signal
23	OPTION SELECT 3	Out	Option Select 3
24	EJECT	Out	Eject Signal
25	EJECT MASK	Out	Eject Mask Signal
26	LED BLINK	Out	LED Blink Signal
27	DISK IN	In	Disk Insert Signal
28	ERR DISK	In	Disk Insert Error Signal
29	FDD INT	In	Disk Interrupt Signal
30	GND	Out	Ground
31	GND	Out	Ground
32	GND	Out	Ground
33	GND	Out	Ground
34	GND	Out	Ground
35	GND	Out	Ground
36	GND	Out	Ground
37	N.C	-	Not Connected

◎ SCSI Device Connection Connector

Pin No.	Signal Name	Pin No.	Signal Name	I/O	Notes
1	GND	26	D0	I/O	Data Signal
2	GND	27	D1	”	”
3	GND	28	D2	”	”
4	GND	29	D3	”	”
5	GND	30	D4	”	”
6	GND	31	D5	”	”
7	GND	32	D6	”	”
8	GND	33	D7	”	”
9	GND	34	DBP	-	” Data Bus Parity Bit
10	GND	35	N.C	-	Ground
11	GND	36	N.C	-	-
12	GND	37	N.C	-	-

Pin	Signal	Pin	Signal		
13	NC	38	TMPWR	Out	Termination circuit use power
14	GND	39	N.C	-	Ground
15	GND	40	N.C	-	Ground
16	GND	41	ATN	-	Attention signal
17	GND	42	N.C	-	Ground
18	GND	43	BUSY	In	Controller active signal
19	GND	44	ACK	Out	Data transfer acknowledgment signal
20	GND	45	RESET	Out	Reset signal
21	GND	46	MSG	In	Command completion response signal
22	GND	47	SEL	Out	Select signal
23	GND	48	C/D	In	Command/data switching signal
24	GND	49	REQ	In	Data transfer request signal
25	GND	50	I/O	In	Input/output switching signal

CZ-634C-TN CZ-644C-TN CZ-644C-TN

10. Main Board

[Main board component-layout diagram]

”11. Main Circuit Diagram (1)”

On the bottom left corner, it says:

”- 75 -”

On the bottom right corner, it says:

”- 78 -”

”12. Main basic wiring diagram (2)”

The numbers at the bottom of the page, ”- 77 -” and ”- 78 -”, probably indicate the page numbers in the document.

”13. Main Basic Circuit Diagram (3)”

”14. Main basic wiring diagram (4)”

”— 81 —”

”— 82 —”

CZ-634C-TN CZ-644C-TN CZ-644C-TN

15. Main Basic Wiring Diagram (5)

[Main board schematic diagram]

16. Control Basic Wiring Diagram

(Schematic diagram — all labels are signal names, component designators, and part numbers in English/alphanumeric.)

”17. Control Board”

18. I/O, FD Connector, SCSI Connector, LED Basic Wiring Diagram

Top right header boxes: CZ-634C-TN CZ-644C-TN

(Schematic diagram — all labels are connector pin names, signal names, component designators, and part numbers in English/alphanumeric.)

19. FD, I/O, SCSI Connector, Power LED, Eject, FD-LED Baseboard

[Image 1]

●FD Connector Baseboard

[Image 2]

●I/O Connector Baseboard

[Image 3]

●SCSI Connector Baseboard

(Note: The ”FD” in the text likely stands for ”Floppy Disk”)

CZ-634C-TN CZ-644C-TN

● Power LED Board

● Eject Board ● FD-LED Board

(PCB component-layout diagrams — all remaining labels are component designators and part values in English/alphanumeric.)

CZ-634C-TN CZ-644C-TN

20. Analog Basic Wiring Diagram

(Schematic diagram labeled ”ANALOG” — all remaining labels are signal names, pin names, component designators, transistor/IC part numbers, and component values in English/alphanumeric.)

— 93 — — 94 —

CZ-634C-TN CZ-644C-TN

21. Analog board

[PCB component layout diagram — Analog board. Component reference designators as silkscreened:]

X7245CE

Capacitors: C401–C418 Resistors: R401–R430 Transistors: Q401–Q409 Diodes: D401–D484 Ferrite beads / fuses: FB400–FB487 Variable resistors: VR401, VR402, VR403 ICs: IC401, IC482 PTC, CE, RM401

Top-left corner: ”22. Power Supply Unit Basic Circuit Diagram”

Below the title: ”The parts marked (triangle mark, square mark) are safety critical parts. When replacing, please use the specified parts at the specified locations to ensure safety and performance.”

(Note: The symbols, ”triangle mark, square mark” are substituted for visual symbols in the actual text.)

”23. Circuit Board”

”KYOSAN”

”AQ1716PA011D”

CZ-634C-TN CZ-644C-TN

24. Keyboard basic wiring diagram

[Keyboard schematic diagram.]

IC1 (microcontroller)

```
XTAL1, XTAL2
RESET
P10–P17, P20–P27, P30–P37 (port lines)
TXD REMOTE, RXD, STBY, RDY
```

IC2 (LS374)

Key matrix: row/column scan lines to key switches.

LED indicators: RED, RED, RED, RED, RED, GREEN, GREEN

Connector: keyboard cable / mouse connector

-101- -102-

”25. Keyboard Component Board”

The numbers at the bottom indicate page numbers.

The text located in the upper center of the image reads:

”CZ-634C-TN CZ-644C-TN CZ-634C-TN CZ-644C-TN”

”26. IC Pin Signal (1)

RH-IX0906CEZZ G/ARRAY (CYNTHIA)

Legend:

- (dashed line) VDD (+5V)
- (solid circle) VSS (GND)

RH-IX1095CEZZ G/ARRAY (VIPS)

Legend:

- (dashed line) VDD (+5V)

- (solid circle) VSS (GND)

- 105 -”

Top left diagram: CZ-634C-TN CZ-644C-TN

RH-IX1604CEZZ G/ARRAY (PEDE/C)

Top pin labels (from left to right): 70, 69, 68, 67, 66, 65, 64, 63, 62, 61, 60, 59, 58, 57, 56, 55, 54, 53, 52, 51

Right pin labels (from top to bottom): 50, 49, 48, 47, 46, 45, 44, 43, 42, 41, 40, 39, 38, 37, 36, 35, 34, 33, 32, 31

Bottom pin labels (from left to right): 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20

Left pin labels (from top to bottom): 30, 29, 28, 27, 26, 25, 24, 23, 22, 21

Pin functions (clockwise from top left, starting at pin 30): DQ0, A1, A2, A3, A4, A5, A6, A7, A8, AG, AD, RDQ, OE2, DQ3, DCLK, VO, VP, DO, DE, DF, LVC2, NC, TV

Top right diagram: RH-IX1603CEZZ MB89352

Pin functions (clockwise from top left, starting at pin ATN): ATN, ACK, REQ, MSG, DBC, SE, IO, CD, RD, SD7, SD6, SD5, SD4, SD3, SD2, SD1, SD0, SOF, A7, A6, A5, A4, A3, A2, A1, A0, GND, VCC, RST, WR, RD, INO, OE2, DQ, DRQC, DACK, REG, ADR, FC, CF, ST, RE, ATR2

Middle right diagram: RH-IXI81CEZZ MN4464S

Pin top (from left to right): VCC, CE2, A8, A9, A10, A11, A12, A3, A4, A2, A3, A4, A5, A6, A7, MD9, MD7, MD4, MD2, MD0, IO0/ID0, IO1/ID1

Bottom right diagram: RH-IX091CEZZ (74ALS244)

Pin functions (clockwise from left top, starting at VCC): VCC, CE2, A8, AO, A1, A2, A3, A4, A5, A6, A7, GND, OE1, IO8, IO7, IO6, IO5, IO4, IO3, IO2, IO0, IO1

Bottom middle diagram: NJM2073D-1 (NJM2073 Dual Power Amplifier)

```
Function:
   1 SS
   2 EO
```

```
3 P1
4 VS1
5 SG
6 OUT2
7 VCC
8 SG
9 OUT1
10 AB
```

CZ-634C-TN CZ-644C-TN

IC Terminal Signals (2)

[IC pin assignment diagrams.]

RH-IX1102CEZZ MSM6258VGS-K

RH-IX1108CEZZ (SN74LS486N)

RH-IX1146CEZZ (SN74ALS05)

RH-IX1096CEZZ G/ARRAY10SC(AN)

RH-IX1094CEZZ G/ARRAY (CATHY)

CZ-634C-TN CZ-644C-TN

RH-IX1093CEZZ G/ARRAY (V/CON)

CZ-634C-TN CZ-644C-TN

27. Set packaging method

[Exploded packaging diagram with item callouts:]

RGB cable TV control cable Protective cover Keyboard Mouse I/O board extraction tool Cushioning material

[Box (upper right):] System disk Dictionary disk Word processor SX-WINDOW A SX-WINDOW B

[Box (lower left):] Function label Key top label Poly bag Instruction manual Service shop guide OS manual Basic manual Utility manual SX-WINDOW manual

Poly bag

Warranty certificate X68000 Exe card Packing case NO. label

28. Procedure for Disassembling the Circuit Board

Please remove the screws and connectors marked with circles in the order of the numbers.

CZ-634C-TN CZ-644C-TN

1. Top case screws
2. Floppy Disk Drive fixing screws
3. Power cable for Floppy Disk Drive
4. Floppy Disk Drive data cable
5. Power Supply Unit fixing screws
6. Power Supply Unit
7. Fan fixing screws
8. CD-ROM Drive fixing screws
9. CD-ROM Drive
10. CD-ROM Drive data cable
11. CD-ROM Drive chassis screws

12. CD-ROM Drive chassis

The document seems to be a disassembly diagram for a computer, specifically for models CZ-634C-TN and CZ-644C-TN.